

An Interpreter Layer for AEC: Cross-Modal Grounding and Ontology-Driven Retrieval

Chia Hui Yen

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College of Fine Arts
Department of School of Architecture
Carnegie Mellon University
Pittsburgh, Pennsylvania 15213

Thesis Committee:

Prof. Joshua Bard, Chair
Prof. Daniel Cardoso Llach, Reader

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The two years at Carnegie Mellon have been both enjoyable and meaningful. The beginning of this journey feels like yesterday.

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Abstract

Architecture, Engineering, and Construction (AEC) projects require rigorous coordination across diverse stakeholders and heterogeneous data formats. While industry standards emphasise digital interoperability (e.g., OpenBIM/IFC), day-to-day coordination still relies on manual interpretive labour: practitioners translate unstructured site evidence, such as photos, chats, and floorplan patches, into model-linked, schema-compliant records. This translation is error-prone, leading to information loss across handoffs, ambiguous evidence-to-element linkages, and non-compliance with regulatory schemas. Consequently, this thesis addresses the following core research question: How can AI act as an interpreter middleware to reliably align unstructured site evidence with project data in AEC workflows? We decompose this challenge into two specific research questions: (RQ1) Multimodal Grounding under Ambiguity: How can multimodal site evidence be reliably grounded to specific digital project elements (e.g., IFC entities) amidst geometric repetition and noisy data? and (RQ2) Ontology-Driven Retrieval: How can domain-specific ontological constraints (e.g., IFC type hierarchies and spatial relationships) transform probabilistic neural outputs into ontology-compliant retrieval results, thereby mitigating hallucination risks?

To answer these, we propose a hierarchical neuro-symbolic architecture, includes a fine-tuned vision-language model extracts typed spatial constraints from mixed site evidence; and a symbolic layer executes a priority-ordered graph traversal over an enriched IFC knowledge graph, built from topology-mined training data, while a Graph-RAG reranker identifies the final ranked candidates.

The thesis contributes an empirically validated framework for neuro-symbolic interpreter layer design in IFC-centric AEC workflows; a reproducible evaluation protocol that separates extraction quality from symbolic retrieval quality; and a concrete analysis highlighting where utilising richer graph fingerprints offers the largest unrealised retrieval gains.

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Chapter 1

Background & Introduction

1.1 Motivation: The Traceability Gap in AEC

1.1.1 Digital Truth vs. Physical Reality

The Architecture, Engineering, and Construction (AEC) industry faces a persistent dichotomy between the structured reality of digital project data and the unstructured nature of physical site execution. On one side, Building Information Modeling (BIM) provides a highly structured, topological representation of a facility, formalized through standards such as Industry Foundation Classes (IFC). Conversely, site evidence is predominantly unstructured and egocentric, captured through ad-hoc methods such as handheld photos, localized text notes, and instant messaging platforms. [13].

1.1.2 The Manual Bottleneck: Semantic Drift and Context Decay

Currently, Construction Managers (CMs) must manually bridge this gap, acting as human translators between the physical site and the digital model. This manual translation introduces two critical vulnerabilities: *Semantic Drift* and *Context Decay*. Semantic Drift occurs when crucial details are lost or misinterpreted during the manual transcription of informal observations into formal records. Context Decay arises when highly localized, idiosyncratic notes degrade over time into unreadable "hieroglyphs," particularly when personnel change or memories fade. Consequently, the **traceability gap** between a physical defect and its digital counterpart is frequently severed.

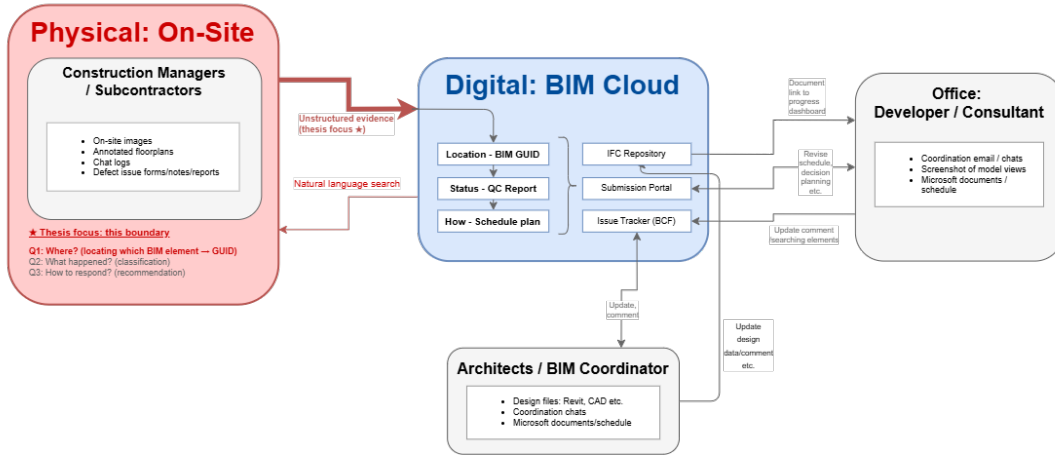


Figure 1.1. AEC workflow and information loss between mutlistakholders and different environments. Image by author.

1.1.3 Vision: Interpreter Middleware and Anchoring to the GUID

The ultimate vision for integrating artificial intelligence into AEC workflows is the deployment of a frictionless Interpreter Middleware. Rather than forcing frontline personnel to adopt rigid software forms or complex tracking hardware, workers should seamlessly submit natural language notes, casual site photos, and chat logs. In a fully realized system, the AI processes this multimodal evidence to automatically answer three core operational questions: ***Where is the issue?***: locating the exact BIM element, ***What happened?***: classifying the status or defect, and ***How to respond?***: reasoning out the next steps.

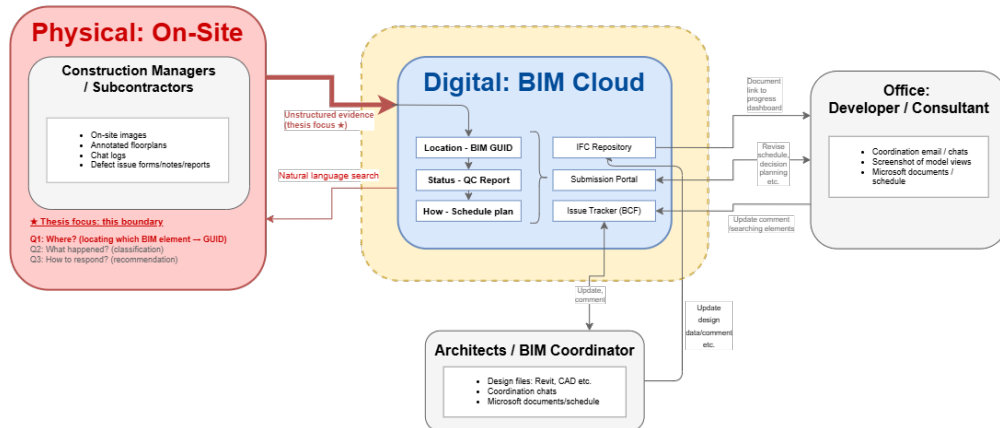


Figure 1.2. Vision, introducing interpreter middleware between workflow. Image by author.

The Localization Bottleneck: However, before an AI can reliably determine what happened or recommend remediation steps, it must resolve the foundational localization bottleneck: exactly *where* is the element? If an AI cannot reliably ground a physical observation to a specific digital entity, subsequent status classifications or schedule updates are rendered unactionable, as they cannot be traced back to the project model. Therefore, solving this foundational "Where" problem from unstructured, multimodal input is the core focus of this thesis.

Why the GUID? To achieve this, the system cannot rely on mere visual or categorical similarity; it must resolve spatial ambiguity down to the exact Global Unique Identifier (GUID). The GUID serves as the absolute, mathematically unique anchor within an IFC model. Anchoring evidence to the GUID is critical for three reasons:

1. **Overcoming Human Ambiguity:** The GUID provides an allocentric, permanent digital pointer that remains invariant.
2. **Machine Interoperability:** Unstructured photos or text notes cannot be directly queried by a computational system. By translating physical observations into a GUID, the evidence is instantly tethered to the model's underlying knowledge graph, unlocking topological properties such as room containment, material definitions, and connected elements. [8]
3. **Regulatory Compliance:** Downstream digital platforms and regulatory schemas, such as Singapore's CORENET X and IFC-SG, mandate strict schema-compliant records. These systems reject localized visual coordinates; regulatory traceability strictly requires that every reported observation be anchored to a verified BIM GUID. [6]

1.2 Current Landscape: Technical Foundations

1.2.1 Structured Data: IFC and Regulatory Schemas

The backbone of AEC digital interoperability is the Industry Foundation Classes (IFC) standard, which provides a rigorous semantic and topological schema for building data. The necessity for strict compliance with such structured data formats is increasingly driven by regulatory frameworks, such as Singapore's CORENET X, which mandate automated, schema-compliant submissions for project approvals.

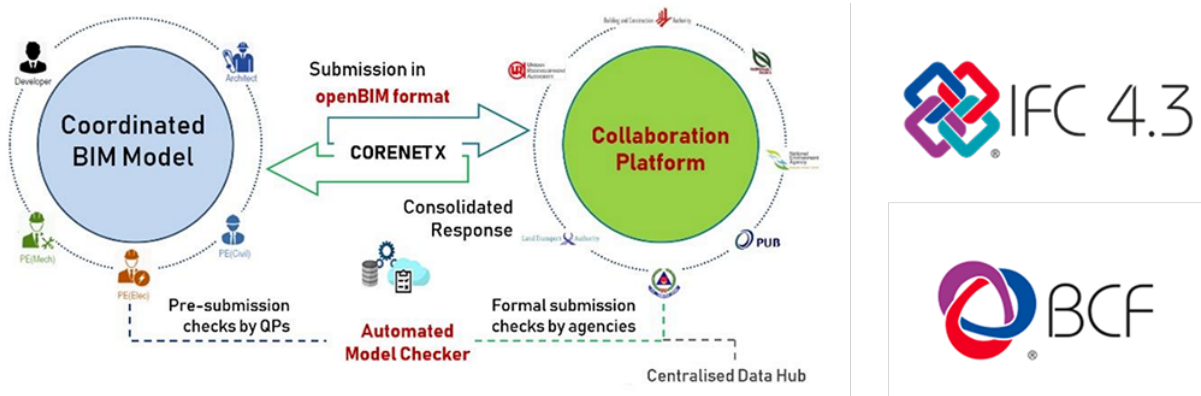


Figure 1.3. Ecosystem of regulatory approval in Singapore, highlighting the role of CORENET X and openBIM formats in unifying submissions [5, 14].

1.2.2 Unstructured Data: The Rise of Multimodal AI

Simultaneously, the rapid advancement of Vision-Language Models (VLMs), such as GPT-4V and Gemini, has enabled machines to process and interpret generic 2D images and unstructured text. These models demonstrate unprecedented capabilities in generic visual recognition and natural language understanding, offering a potential mechanism to parse the messy reality of construction sites [11].

Within the AEC context, these capabilities have been applied to design assets generating, information retrieval and semantic enrichment by extracting and synthesizing information. Recent frameworks, such as *NADIA* [28] and *Text2BIM* [18], demonstrate the capacity for Large Language Models (LLMs) to serve as design assistants by translating natural language instructions into software-executable code (e.g., Python scripts for Revit or Rhino).

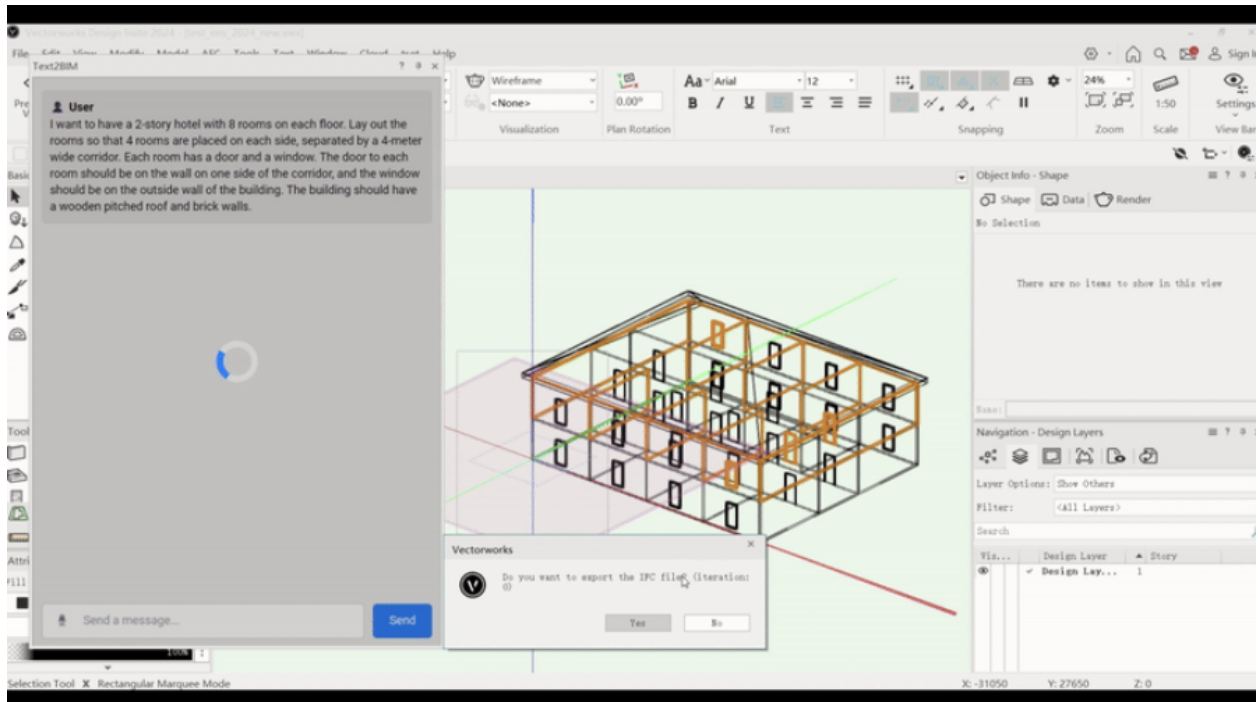


Figure 1.4. Example of Text2BIM-style workflows where design intent in natural language is translated into BIM geometry [18].

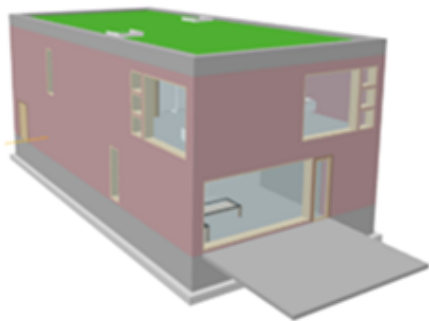


Figure 4: Duplex House

Q: Which rooms have direct access to outdoor spaces?
A: [...] 4 Rooms have direct access to outdoor spaces: A101, B101, A102, B102. [...]
Q: Are the walls of room A203 load bearing?
A: [...] The interior partition and plumbing walls in room A203 are NOT load-bearing [...]
Q: What is the depth of the foundation?
A: [...] The depth of the foundation is 0.3 meters (300 mm). [...]
Q: Are all internal doors compliant with accessibility requirements?
A: Based on my analysis of all 14 doors in your IFC project, not all internal doors are compliant with accessibility requirements. [...]

Figure 5: Questions and correctly given IFC-based answers by Sonnet 4.5 on the ARC Duplex using MCP4IFC.

Figure 1.5. Example of current LLM-based interactions with BIM models (MCP4IFC), showing Q&A capabilities but highlighting the gap in handling downstream regulatory schemas [34].

1.3 Core Gaps in the Current Landscape

1.3.1 The Visual vs. Geometric Semantics Gap

Despite the promise of VLMs, a fundamental disconnect persists between how AI perceives a scene and how a building is logically structured—a gap between the visual and geometric semantics logic, "Eyes" and the "Brain." While Vision-Language Models (VLMs) process all visual inputs natively as uniform 2D pixel arrays, this thesis treats on-site photographs and architectural floorplans as distinct semantic modalities due to extreme representation heterogeneity. Photographs provide an egocentric, photometric representation, requiring the interpretation of perspective, texture, and occlusion. Conversely, floorplans offer an allocentric, symbolic representation defined by orthographic geometry and absolute spatial relationships. Effective multimodal grounding must synthesize these diverging visual logics alongside natural language. Because VLMs are fundamentally blind to underlying engineering topologies, such as connection pattern and spatial relationships (e.g., `IfcRelConnects`), mapping these heterogeneous visual observations directly to strict digital models predictably induces hallucinations.

To be specific, here are three specific gaps explain why a general-purpose multimodal LLM is insufficient for AEC interpretation:

- **The Linearization Mismatch:** Current LLM-BIM integrations often rely on linearizing IFC data into text tokens. As noted by Iranmanesh et al. ([26]), this discards the explicit relational structure of the **Nested Node Pattern**, forcing the model to reconstruct construction topology implicitly. This motivates a graph-aware approach where IFC is parsed into an entity-relationship graph first, then queried through an agentic tool-use API [34, 45].
- **Domain Specificity:** Generic models (e.g., GPT-4) lack an understanding of construction-specific details, such as joints or reinforcement dependencies. These domain-specialized tools need to be orchestrated by an agentic layer rather than being used "out of the box" [12, 31].
- **Generation vs. Interpretation:** Most existing literature focuses on *producing* geometry (e.g., *Text2BIM*) but skips the critical problem of mapping messy as-built evidence back to regulatory schemas with zero-hallucination [18, 27].

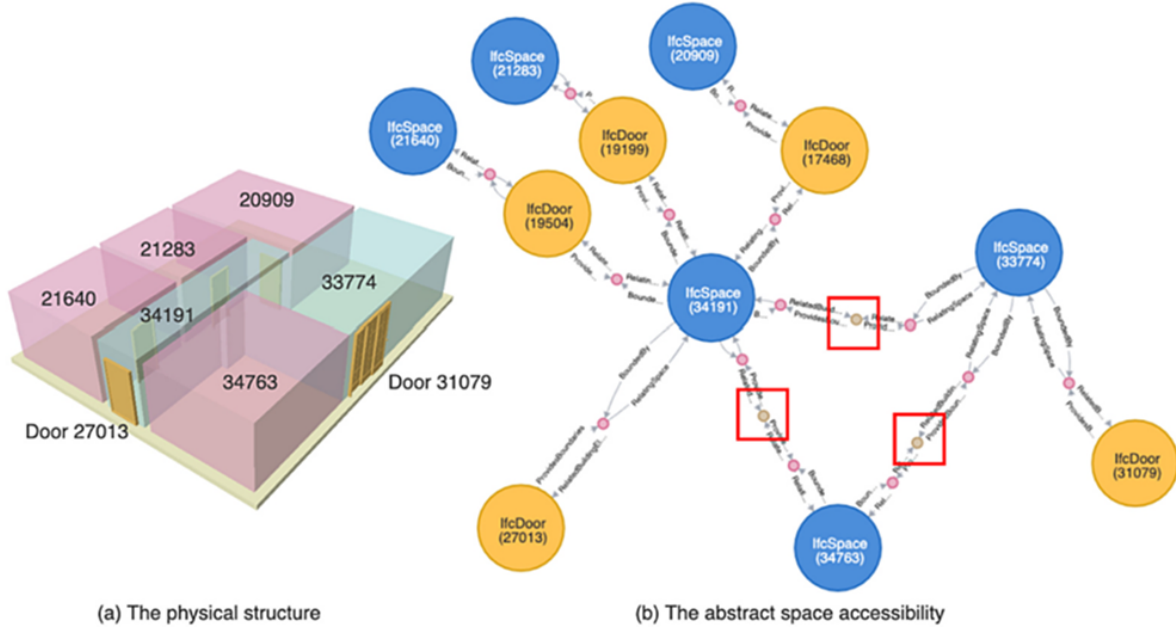


Figure 1.6. Comparison between (a) the physical structure of a building module and (b) its abstract graph representation (IfcSpace, IfcDoor) used for topological reasoning [45].

1.3.2 The Mathematical Deadlock: Attribute Entropy

This topological blindness leads to a mathematical deadlock, particularly in high-repetition environments like Prefabricated Prefinished Volumetric Construction (PPVC). In such projects, a single floor might contain dozens of identical elements (e.g., 46 identical windows sharing the exact same geometric and semantic attributes). In these scenarios, pure attribute filtering fails entirely, yielding baseline accuracy rates as low as 2.2%, 1 over 46. Pure visual AI cannot break this ambiguity deadlock without integrating the spatial topology of the building graph.

1.4 Research Questions and Scope

1.4.1 Core Research Questions

To address these gaps, this research investigates the following overarching question:

How can AI act as an interpreter middleware to reliably align unstructured site evidence with project data in AEC workflows?

To address this, we decompose the problem into two specific sub-questions that directly respond to the gaps identified in Section 1.3.

1. **RQ1: Multimodal Grounding under Ambiguity.** *How can multimodal site evidence be reliably grounded to specific digital project elements (e.g., IFC entities) amidst geometric repetition and noisy data?*

Rationale: This question directly addresses the visual versus geometric semantics gap (Section 1.3). Because foundation models are trained on generic, open-domain data, they lack an inherent understanding of AEC-specific topologies. Answering this question establishes how multimodal triangulation—combining unstructured visual cues with spatial topology cues extracted—can successfully bridge the divide between ego-centric site photos and allocentric digital entities.

2. **RQ2: Ontology-driven Retrieval.** *How can domain-specific ontological constraints (e.g., IFC type hierarchies and spatial relationships) transform probabilistic neural outputs into deterministic retrieval results to mitigate hallucination?*

Rationale: This question addresses the operational requirement for strict determinism in AEC workflows. Visual perception alone is insufficient for engineering validity; predictions must be constrained by the actual topological structure of the building graph (e.g., spatial containment, element connectivity). Answering this question determines how a neuro-symbolic architecture can successfully shift AI operations from generating plausible (but potentially hallucinated) text toward the verifiable retrieval of exact IFC elements.

1.4.2 Scope and Delimitations

This research focuses specifically on the interoperability challenges inherent in industrialized construction, utilizing Singapore’s **PPVC (Prefabricated Prefinished Volumetric Construction)** ecosystem as the primary context. The extreme geometric repetition of PPVC buildings provides an ideal stress test for AI disambiguation capabilities. Due to the privacy constraints of live-project data and the scarcity of open-source high-rise PPVC datasets, the proposed system is evaluated using a controlled synthetic sandbox. These synthetic datasets are derived from general IFC standards and engineered to simulate the topological and temporal features of mass housing workflows, ensuring rigorous validation of the proposed neuro-symbolic pipeline that we will introduce in Chapter 4.

Chapter 2

Related Work

This chapter situates the proposed research within the intersection of industrialised construction, semantic interoperability, and multimodal machine learning. The review is structured to build a theoretical foundation for a **Hierarchical Neuro-Symbolic Architecture** supported by probabilistic multimodal perception, deterministic query planning, and topological graph retrieval.

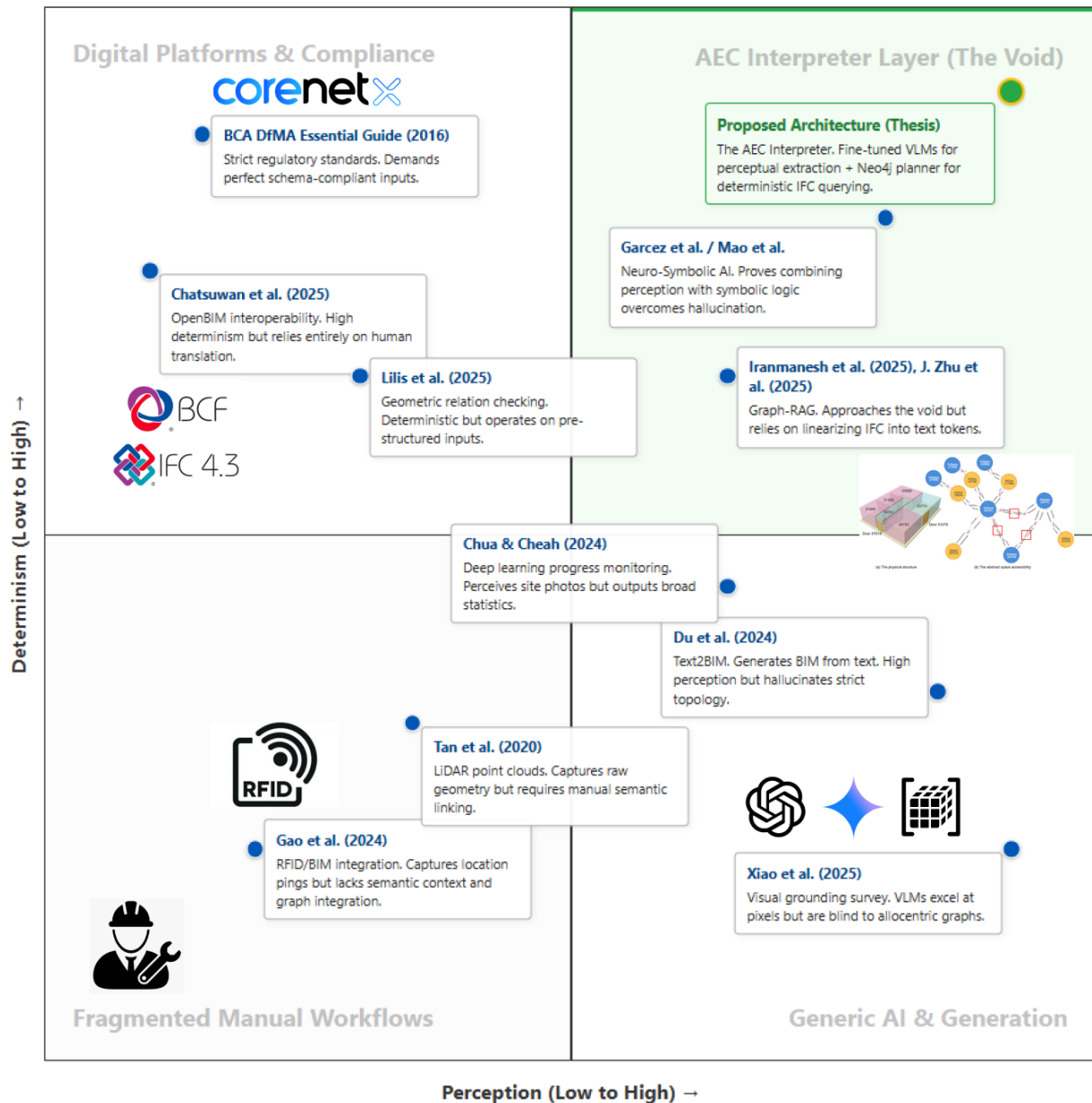


Figure 2.1. The Interoperability Matrix: Exposing the gap between probabilistic neural perception (Generative AI) and deterministic symbolic representation (IFC/BIM) in the AEC execution phase. Image by author.

2.1 Digital Continuity in Industrialized Construction

The AEC industry is undergoing a structural transition toward Industrialized Construction, characterized by the adoption of Design for Manufacturing and Assembly (DfMA) principles to enhance precision and productivity. Within this domain, Prefabricated Prefinished Volu-

metric Construction (PPVC) serves as a limit case for extreme standardization. In markets such as Singapore and Malaysia, PPVC is increasingly mandated to enforce rigorous quality control in high-density urban developments [6, 42].

2.1.1 The Information Friction in Site-to-Model Alignment

The efficacy of these industrialized workflows is predicated on digital continuity—the seamless flow of information between physical site execution and the underlying digital twin. However, as building components move from fabrication to assembly, the link between physical truth and digital records remains a significant source of information friction [6]. This friction results in a divergence where the ”as-built” reality frequently drifts from the ”as-designed” intent, leading to costly rework and regulatory non-compliance.

2.1.2 Limitations of Hardware-Centric Monitoring

Current efforts to maintain digital continuity rely heavily on automated data acquisition hardware. These are generally categorized into sensor-based tracking (RFID/UWB) [23] and reality capture (Laser Scanning/Photogrammetry) [1]. However, these approaches encounter significant operational bottlenecks:

- **The Perception-Interpretation Gap:** While hardware can capture the existence of an object, it lacks the logic to interpret its semantic state. This leads to a data silo, where large volumes of raw sensor data or point clouds are collected but remain disconnected from the BIM’s semantic graph because they lack a logical bridge to the project’s ontology.
- **Environmental Occlusion and High Entropy:** In repetitive high-rise environments, signal attenuation and ”dynamic occlusions” (scaffolding, site activity) mask the build state [13]. Furthermore, in states of high Attribute Entropy—where modules are geometrically identical—hardware often fails to ground a physical observation to a unique digital entity (the ”Mathematical Deadlock” identified in Section 1.5).

2.1.3 The Need for a Streamlined Interpretive Workflow

The limitations of hardware-centric methods suggest that resolving digital continuity is not merely a problem of more data acquisition, but of data pipeline orchestration. There is a clear requirement for a computational logic that can integrate heterogeneous site data, from low-cost inputs such as smartphone photos, into structured digital outputs.

Specifically, the literature points toward the necessity of an interpretive mechanism that can:

1. Resolve spatial ambiguity in high-repetition environments by leveraging topological priors rather than purely visual signatures.
2. Align messy, unstructured site evidence with the structured representations defined by IFC ontologies (e.g., element types, spatial containment, connectivity).
3. Bridge the gap between physical data acquisition and digital compliance through a middleware that organizes raw site observations into structured, traceable records.

This identifies a research opportunity: the development of a software-based logic that moves beyond perception toward verifiable alignment across the AEC data pipeline.

2.2 Semantic Interoperability: From Static Schemas to Knowledge Graphs

To resolve the friction between egocentric site evidence and allocentric project records, a system must operate within a robust semantic framework. This section evaluates the transition of the **Industry Foundation Classes (IFC)** from a medium of data exchange to a graph-based foundation for machine reasoning.

2.2.1 The IFC Standard as Semantic Backbone

The backbone of digital construction is the **Industry Foundation Classes (IFC)** standard. As the only globally recognized open standard for BIM data exchange, IFC provides the necessary platform independence for interoperable AI systems [35]. Its primary architecture, the **EXPRESS schema (ISO 10303-21)**, serves as a rigid and reliable format designed for cross-platform file exchange. This rigidity ensures that regardless of the authoring tool used, the "digital truth" of the design remains consistent across the project lifecycle.

Despite its strength in data exchange, the EXPRESS-based hierarchy presents challenges for AI-driven interpretation. The format is **computationally expensive to traverse**, particularly when resolving complex element connectivities or nested spatial relationships [10]. The linear and deeply nested nature of traditional IFC files makes it difficult for an agent to perform the multi-hop reasoning required to disambiguate identical elements.

From Linear Files to Building Knowledge Graphs (KGs) To unlock the potential of AI in AEC, the literature identifies a shift toward more flexible data representations, such as **ifcJSON** and **Linked Data**. These formats allow the building model to be treated as a **Labeled Property Graph (LPG)**.

Representing IFC as a graph transforms the building model into a searchable knowledge base. This evolution enables **LLM-assisted Information Extraction**, where a foundation model acts as a parser to extract properties and relationships from the IFC data [26]. Unlike traditional methods that treat the file as a static document, a graph-based approach allows for **Graph-RAG** (Graph Retrieval-Augmented Generation). In this paradigm, the system does not merely search for keywords; it traverses the topological “neighbourhood” of an element—querying its material layers, spatial containers, and connected neighbours—to constrain probabilistic visual guesses toward structured engineering records.

2.2.2 IFC-SG and Regulatory Compliance

In Singapore, the emergence of the **IFC-SG** standard further underscores the need for this semantic precision. IFC-SG is designed for automated regulatory checking and requires that submitted BIM data conform to specific, machine-interpretable fields [7]. This regulatory direction motivates the present work: as the industry moves toward such schema-driven submissions, there is a growing need for automated mechanisms that can bridge unstructured site data and structured BIM references. This thesis studies the IFC-SG workflow to understand what downstream fields matter, but the prototype focuses on the upstream GUID retrieval step rather than full IFC-SG compliance [8].

2.3 Multimodal Perception and Visual Grounding in AEC

This section evaluates the transition from generative models toward interpretive frameworks, focusing on the ability of Vision-Language Models (VLMs) to ground visual evidence within the rigid constraints of architectural logic.

2.3.1 From Generative Design to Autonomous Interpretation

Recent literature in the AEC domain shows a shift from static generative models toward agentic frameworks that leverage foundation models for task orchestration.

- **Generative Pipelines:** Generative-intent driven implementations such as **Text2BIM** [18] and **Text2CAD** [40] demonstrate the capacity for LLMs to transform natural language into imperative code for geometry creation. However, these systems operate in a “forward-loop,” assuming a clean-slate design environment rather than addressing the messy, retroactive alignment of site execution data.

- **Autonomous BIM Agents:** The emergence of "Computer-Use Agents" marks a transition toward interaction. **BIMgent** [17] utilizes a vision-driven approach to automate modeling actions through Graphical User Interfaces (GUIs), effectively mimicking human software interaction. Similarly, **MCP4IFC** [34] utilizes the Model Context Protocol to bridge LLMs with BIM environments, allowing for context-aware model editing.
- **Scene Understanding:** Frameworks like **SceneScript**[2] and **MeshCoder** [15] have advanced the "Perception-to-Code" pipeline, utilizing point clouds and scene encodings to predict structured geometry.

2.3.2 The Semantic Gap: Visual Perception vs. Geometric Logic

Despite these advances, a gap persists between visual recognition and engineering reasoning. AI models can identify building elements from images, but they cannot infer the typed relationships (e.g., containment, connectivity) that define those elements in an IFC graph.

- **Pixel-Bound Perception:** Foundation models like CLIP [37] and GPT-4o perceive architectural data as 2D pixel arrays. While they excel at identifying a "wall" or "window" based on texture and contrast, they are blind to the underlying engineering graph.
- **Reconstruction Limitations:** Current **Img2CAD** approaches [44] attempt to fill geometric parameters from single-view images using flow-matching techniques. However, these reconstructions often focus on visual parameters (attributes) rather than the structural relationships (e.g., `IfcRelConnects`) required for compliance.
- **Spatial Predicate Deficiency:** As highlighted by [Xiao et al. \(2025\)](#), generic pixel-based visual grounding models often struggle to disambiguate targets in complex scenes without explicit spatial prior relations (e.g., recognizing that object A is "to the left of" object B). Translated into the context of architectural reasoning, this indicates a critical deficiency: purely visual models lack the strict topological Spatial Predicates (e.g., `IfcRelFillsElement`, `ADJACENT_TO`, `CONTINUOUS`) required by AEC schemas. Without bridging this gap to include topological predicates, an agent cannot differentiate between a generic visual category (a "window") and a specific, uniquely identified digital entity (its GUID) within a highly repetitive environment.

2.3.3 Visual Grounding and Ambiguity in AEC

Visual grounding—the process of mapping a natural language query to a specific region or instance in an image—is the technical core of site-to-model alignment.

- **Scene Graph Generation (SGG):** Recent research into VLM-based SGG [39] attempts to map visual tokens to industrial scene graphs. However, these graphs often remain at a high level of abstraction and fail to map to the granular, nested hierarchies of a BIM.
- **Grounding under High Attribute Entropy:** Current techniques assume unique targets within a scene. In industrial high-rise buildings (e.g., PPVC), where Attribute Entropy is high, visual signatures are non-unique.

2.3.4 Toward Ontology-Constrained Neural Models

To resolve these failures, recent literature suggests that neural models must be constrained by symbolic ontologies.

- **Intent Recognition:** Research such as **BIM Command Recommendation** [19] uses Transformers to predict user intent from logs. Expanding this to site-to-model interpretation requires aligning neural perception with the project’s semantic goals.
- **Interactive Detailing:** Systems like **NADIA** [27] demonstrate the potential for natural language to drive architectural detailing through interaction. This suggests a shift toward AI as a mediator that interprets site evidence to “fill” the requirements of a BIM schema.
- **Ontology-Aligned Parsing:** To prevent hallucinated outputs that violate the building ontology, frameworks are moving toward Auto-Cypher generation [38] and Graph-RAG [26]. In these cases, the neural model generates queries rather than final outputs, allowing a deterministic graph database to verify site evidence against the project’s ontology.

2.3.5 Comparative Summary

Table [Table 2.1](#) summarizes the computational distinctions between existing neural perception systems and the requirements for an interpretive data pipeline.

As Table [Table 2.1](#) shows, several systems incorporate grounding mechanisms (rule-based checkers, RAG-based search, geometric constraints). However, none of them address the

Table 2.1. Comparison of AI Mechanisms for Architectural Representation and Interpretation

System	Interpretation (Input → Context)	Grounding Mechanism
<i>Agentic Tools</i>		
Text2BIM	Text → Multi-agent design planning	Rule-based model checker.
MCP4IFC	Text + Scene Metadata → Tool Intent	RAG-based graph search.
BIMgent	Visual GUI state → Software Plan	None (Pixel-bound automation).
<i>Representational</i>		
MeshCoder	Point Cloud → Shape Tokens	Geometric (CD). Chamfer Distance
Text2CAD	Textual Intent → Latent Embedding	CAD Sketch/Extrude Constraints.
SceneScript	Video/Cloud → Scene Encoding	None (Statistical sequence).
Img2CAD	Image → Parameter Flow Matching	Attribute-specific matching.
Proposed	Heterogeneous Site Evidence	Deterministic IFC Graph

specific challenge of disambiguating visually identical elements in high-entropy environments using topological predicates from the building graph. Resolving site-to-model alignment in such environments requires a framework that uses the neural model as a front-end for a graph-based verification layer.

2.4 Neuro-Symbolic Architecture

To resolve the stochastic baseline failure identified in Section 2.3, the literature suggests a transition from purely neural perception toward Neuro-Symbolic (NeSy) integration [16, 33]. In this paradigm, a probabilistic neural layer processes noisy, multimodal site evidence, while a deterministic symbolic layer enforces logical consistency within the project’s ontology.

2.4.1 Neuro-Symbolic Integration: Bridging Perception and Logic

Drawing from the principles of Neuro-Symbolic AI, an interpretive framework for AEC must move beyond pattern matching alone. The architecture is defined by the depth of complementarity between layers:

- **The Neural Layer:** A domain-adapted Vision-Language Model (e.g., via LoRA fine-tuning) maps site evidence (photos, text notes) into structured constraint fields such as element type, storey, and spatial predicates.
- **The Symbolic Layer:** Responsible for enforcing engineering constraints. It utilizes the IFC graph to verify and ground the neural layer’s outputs against the project’s topological structure and spatial predicates.

In this configuration, the neural model acts as a query generator: it transforms unstructured site data into formal instructions, such as Cypher queries, that allow the symbolic layer to perform deterministic retrieval against the building knowledge graph.

2.4.2 Topological Complexity and the Nested Node Pattern

A primary challenge in symbolic grounding is the inherent topological complexity of the building graph. [Iranmanesh et al.](#) identifies the Nested Node Pattern within IFC data as a significant barrier for standard AI models.

- **Path Complexity:** In the IFC graph, simple spatial relationships (e.g., an `IfcWindow` contained within an `IfcWall` on an `IfcSpatialElement`) involve multiple levels of hierarchical nesting and implicit associations.
- **The Requirement for Multi-hop Reasoning:** Standard Retrieval-Augmented Generation (RAG) fails to navigate these deep topological dependencies. Literature indicates that a graph-aware approach, utilizing multi-hop query templates (e.g., Cypher traversals), is necessary to prevent hallucinated relationships between entities. In this approach, the system verifies spatial predicates against the actual IFC graph before returning candidate elements, rather than relying on the neural model alone to produce correct identifiers.

2.4.3 Summary of the Research Gap

The review reveals a mature landscape in generative BIM geometry and static architectural graph querying. However, a research gap remains regarding the automated interpretation

of unstructured site evidence for element-level retrieval. Existing approaches fall into two paradigms that have not been combined for this purpose:

- **Probabilistic Neural Perception:** Current foundation models excel at visual pattern recognition but operate without topological logic. In high-entropy construction environments, this results in the stochastic baseline failure described in Section 1.3.2.
- **Deterministic Symbolic Representation:** IFC-based graphs encode engineering truth but lack the flexibility to ingest unstructured, heterogeneous real-world data without human intervention.

This thesis applies existing techniques from both paradigms, specifically Parameter-Efficient Fine-Tuning (PEFT/LoRA) for probabilistic, domain-adapted constraint extraction and graph traversal (Cypher queries over Neo4j) for deterministic topological disambiguation to address the site-to-model retrieval problem in high-entropy AEC environments.

Chapter 3

Research Method

3.1 Overview

This thesis adopts a **Constructive Design Research (CDR)** approach. Rather than treating AEC data purely as a static dataset for analysis, this research treats the high-rise building construction workflow, take PPVC as main example as a *field site* in which a **neuro-symbolic retrieval system** is introduced, iterated, and reflected upon.

The core premise is that the act of constructing the prototype is itself the primary mode of inquiry. Building a system that must retrieve specific IFC elements from a BIM model using unstructured site evidence exposes semantic gaps that purely theoretical models overlook. The workflow studied is informed by regulatory frameworks such as CORENET X, which establish the downstream need for GUID-level traceability.

3.2 Constructive Design Research (CDR) as an Approach

3.2.1 Why CDR?

Constructive Design Research refers to design research in which the construction of an artefact is the primary means of generating knowledge [29]. It is well-suited for this thesis because interoperability in AEC resists both pure scientific inquiry (which requires controlled variables often absent in construction) and pure engineering (which focuses on optimization rather than meaning-making).

Following Frayling’s [22] “Research through Design” framing, this thesis validates its hypotheses by constructing an artefact and observing how it performs against real data.

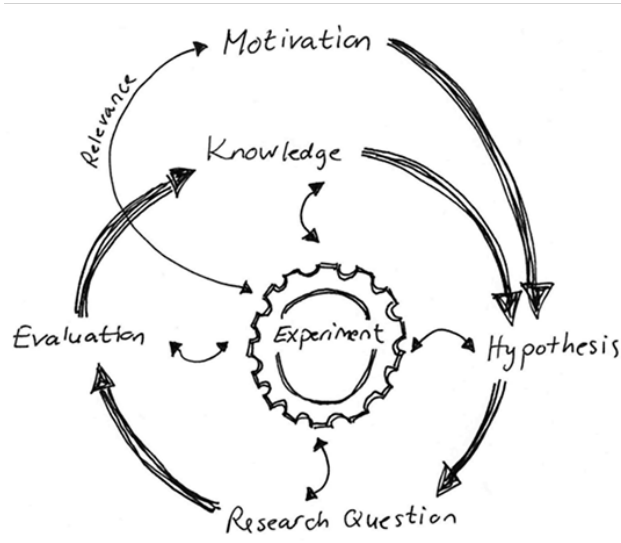


Figure 3.1. Constructive Design Research (CDR) as articulated by Bang et al. [3], where experiment serves as the driving wheel of knowledge generation. In this thesis, the interpreter prototype functions as the core experiment for testing and refining the hypothesis about an AI-assisted interpreter middleware in AEC workflows.

3.2.2 Methodological Guidelines: Ensuring Intelligibility

To ensure rigor, we follow the **Multiple Intelligibility** framework of Lee et al. [30], which requires the research to be legible to diverse audiences:

- **Design-relevant methods:** The prototype operates on standard IFC models and queries the building graph through established database tools (e.g., Neo4j), keeping the system intelligible to AEC professionals [4, 5].
- **Theoretical base:** The architecture follows a neuro-symbolic design [16]: a domain-adapted neural model extracts structured constraints, while a symbolic layer performs deterministic retrieval against the building graph.
- **Recipient design:** The dataset and evaluation are informed by PPVC workflows and IFC-SG guidelines [4, 5, 7], ensuring the evaluation is relevant to the regulatory ecosystem [30].
- **Dialogic process:** The synthetic dataset curation makes the gap between raw site evidence and machine-readable constraints explicit, revealing where the system succeeds and where it fails.

Within the spectrum of CDR, this thesis adopts a pragmatic, lab-based prototyping approach [20]. The central activity of this research is the construction of a **neuro-symbolic interpreter middleware**, which functions as a “research probe” to operationalize theoretical gaps into testable software tasks.

3.3 Research Design and Prototype Overview

Getting back to our research, the ultimate aim of this prototype is to operationalize the research questions (Section 1.4) into testable tasks:

1. **Addressing RQ1: Multimodal Grounding under Ambiguity.** The prototype evaluates whether multimodal evidence can be grounded to specific IFC elements amidst geometric repetition, using topological relationships such as adjacency and connectivity rather than visual matching alone.
2. **Addressing RQ2: Ontology and Schema Alignment.** The prototype tests whether ontological constraints from the building graph can transform probabilistic neural outputs into verifiable retrieval results.

3.3.1 The Neuro-Symbolic Hypothesis: Balancing Flexibility and Determinism

To address the current workflow traceability gap identified in Chapter 1.1.2, contextualized in Chapter 1.3, supported by current landscape mentioned in Chapter 2.4.3, this research operates on the “**Neuro-Symbolic Hypothesis.**” This hypothesis asserts that neither purely neural nor purely symbolic approaches are individually sufficient for AEC retrieval tasks. Instead, decoupling the system achieves an optimal balance:

1. **Flexibility vs. Strict Compliance:** Foundational Vision-Language Models (VLMs) excel at generic generation over messy human inputs but lack the strict topological determinism required by AEC regulations. Conversely, traditional symbolic systems (e.g., Cypher) are perfectly deterministic but fail when inputs are ambiguous. The neuro-symbolic approach bridges this by forcing the VLM to generate only intermediate constraints, allowing the symbolic planner to ground these fuzzy intents into strict IFC compliance.
2. **Computational Efficiency:** Asking an LLM to perform linear, typed-based token vector database or raw graph traversals “in-context” leads to severe token limits and high latency. Offloading the heavy topological traversal to an optimized graph database drastically reduces inference latency.

3. **Attribute Entropy Mitigation:** High-rise building environments suffer from extreme attribute entropy. Deferring final disambiguation to the symbolic graph index leverages exact spatial coordinates and relationships that are invisible to the neural model.

3.3.2 The Sandbox: Synthetic Data as Field Proxy

Due to data privacy constraints in the AEC industry, the prototype cannot be evaluated on live project data. Instead, a controlled evaluation environment (a “sandbox”) is constructed using semi-synthetic data derived from open IFC models. The dataset is designed to reproduce the key challenges of PPVC projects, specifically high geometric repetition and fragmented multimodal evidence, allowing controlled evaluation in the absence of a live field site. The data curation process is described in detail in Chapter 4.

Detailed implementation of these architectural components is presented in Chapter 6, while the experimental protocols used to evaluate them within the sandbox are presented in Chapter 5.

Chapter 4

Data Curation and Ground-Truth Definition

This chapter establishes the data foundation required to build and evaluate the proposed neuro-symbolic interpreter middleware.

The chapter is organized into three primary sections:

- **Ground-Truth Definition:** Establishes the dual-layer baseline anchoring the study, comprising the authoritative IFC geometric truth and the schema-compliant structured truth.
- **Synthetic Data Curation:** Details the pipeline used to transform raw IFC semantics into paired multimodal test cases while preserving deterministic traceability.
- **Data Limitations:** Addresses the physical constraints of native IFC topology and evaluates the ecological validity of the synthetic dataset.

4.1 Core Data Assets and Ground-Truth Definition

The evaluation framework in this thesis operates on the principle that the intermediate neuro-symbolic reasoning and the final element retrieval must be objectively measurable. To address the core research questions, **multimodal grounding under ambiguity (RQ1)** and **ontology-aligned constraint extraction (RQ2)**, the system must map probabilistic multimodal evidence to deterministic architectural elements, from unstructured on-site data evidence to exact Global Unique Identifier (GUID) of the target element as Chapter [3.3.1](#) and [1.1.3](#) mentioned.

Therefore, the primary ground truth in this study is *geometric and topological* (Section [4.1.1](#)). The IFC model acts as the absolute reference not only for the final retrieved element (evaluating RQ1) but also for the spatial relationships, subtypes, and directional constraints

extracted by the neural layer (evaluating RQ2). By treating the IFC graph as the ground truth and ontology database for retrieval database and evaluation.

4.1.1 Geometric Ground Truth: The IFC Models

The first layer of ground truth in this study is geometric and topological. Each evaluation case must ultimately resolve to a specific IFC entity whose identity, spatial containment, and neighbouring relations can be checked deterministically against the source model. For this reason, the evaluation environment is anchored in three Industry Foundation Classes (IFC) models, which function not merely as background reference files but as the authoritative source of element identity and building structure. This design follows the graph-oriented interpretation of IFC discussed in Chapters 2 and 3, where the building model acts as a structured substrate for retrieval and verification rather than as a passive geometric repository [35, 45].

The three IFC models are used with different methodological roles. `AdvancedProject.ifc` serves as the primary evaluation model because it contains the density, repetition, and vertical hierarchy required to test ambiguous site-to-model grounding under realistic retrieval pressure. `BasicHouse.ifc` and `Duplex_A.ifc` are included as smaller cross-IFC comparison models in order to check whether the retrieval logic and curation assumptions remain meaningful outside the primary file. Together, these models support both the main ambiguity experiments and the later robustness checks across different IFC structures.

- **Model A (AdvancedProject):** a complex multi-storey model used for primary retrieval and topology evaluation.
- **Model B (BasicHouse):** a compact residential model used for cross-IFC robustness checks.
- **Model C (Duplex_A):** a split-level residential model used for cross-IFC robustness checks.

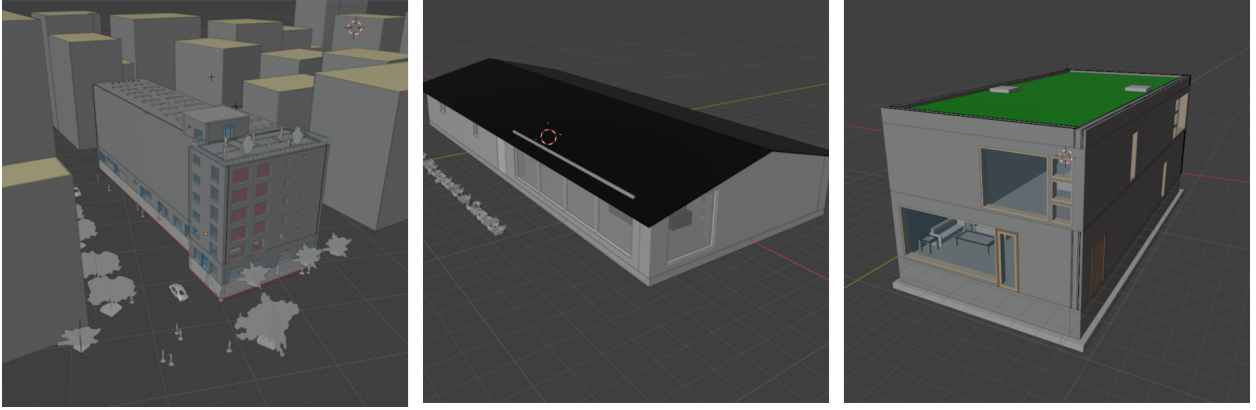


Figure 4.1. Visual representation of the three IFC models used as geometric ground truth. From left to right: Model A (AdvancedProject), Model B (BasicHouse), and Model C (Duplex_A).

Table [Table 4.1](#) summarises `AdvancedProject.ifc`. Its high-density repetition (46 identical windows per floor) and deep spatial structure make it suitable for testing RQ1: correct retrieval cannot rely on coarse attributes alone but must exploit spatial and topological constraints.

Metric	Value	Challenge Implication
Total Indexed Elements	1,233	Large search space for Retrieval (RQ1)
Storey Count	10 (9 with elements)	Vertical ambiguity (e.g., "Upper floor")
Window Count	263 (46 per typical floor)	High geometric repetition (Hard Negatives)
Wall Count	771 (IfcWall + IfcWallStandardCase)	Dominant element class, subtype overlap
Deepest Hierarchy	5 Levels	Requires multi-hop topological reasoning (RQ1)

Table 4.1. Complexity metrics of the primary Geometric Ground Truth (`AdvancedProject.ifc`).

4.2 Synthetic Dataset Curation Strategy

To clearly distinguish between data curation and model training, this thesis refers to the early template-based dataset as the **Baseline Synthetic Dataset (SD-Baseline)**, and the

final LLM-judged, region-anchored dataset as the **Region-Grounded Synthetic Dataset (SD-Grounded)**.

The synthetic dataset in this thesis functions as an evaluation instrument rather than a general substitute for real construction data. **SD-Baseline** contained only 37 unique spatial-relation output templates across 1,064 samples, which inadvertently caused the neural model to learn template shortcuts rather than genuine topology-based reasoning.

To address this, next dataset, **SD-Grounded** introduced a substantially revised curation pipeline. This updated pipeline adopts region-grounded supervision, anchoring training labels directly to IFC-mined wall regions, and applies Large Language Model (LLM)-assisted skeleton review to ensure each case provides a discriminative spatial signal.

SD-Grounded is strictly derived from `AdvancedProject.ifc` (AP). All 297 retained skeletons are drawn from this model and split into a 237-case canonical training set and a 60-case held-out evaluation set, supplemented by 753 augmented training variants for the full training run. **Figure 4.2** illustrates the dataset composition, relation multiplicity, topology family distribution, and augmentation axes for this SD-Grounded assembly.

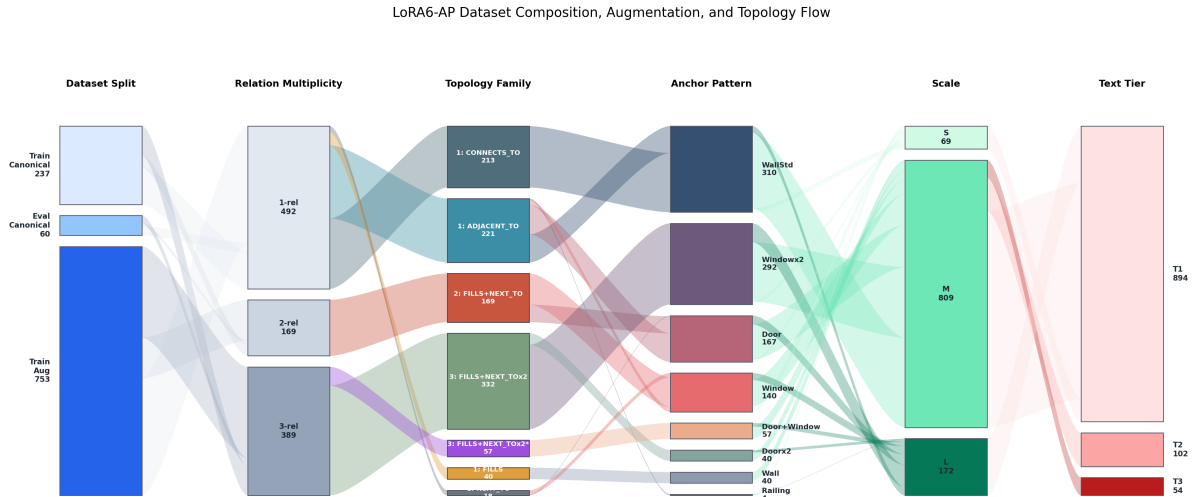


Figure 4.2. SD-Grounded dataset composition, augmentation, and topology flow. The alluvial diagram traces each case from dataset split through relation multiplicity on flat relationship (1-rel: 492, 2-rel: 489, 3-rel: 249), topology family, anchor-element pattern, evidence scale, and text tier. The dominant 3-rel family is `FILLS+NEXT_TO+NEXT_TO` (a window anchored between two openings on the same wall).

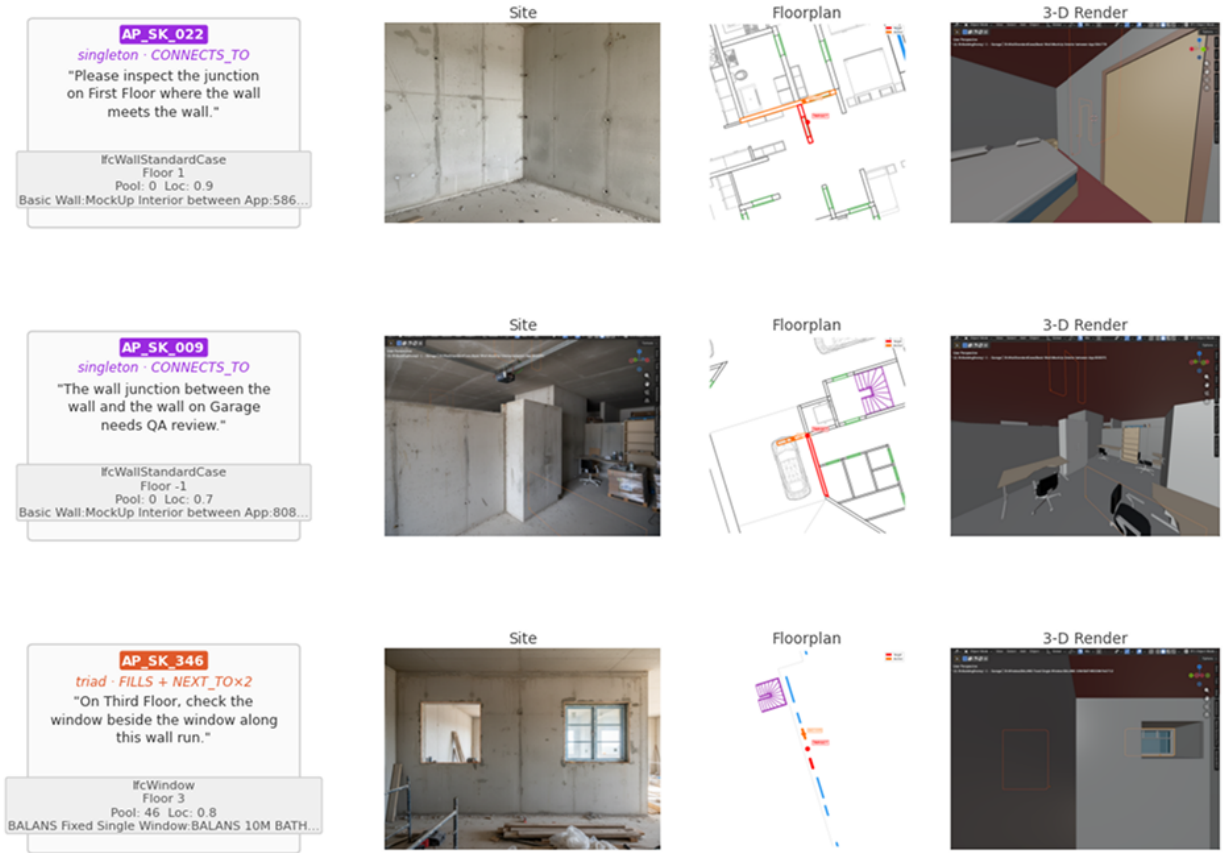


Figure 4.3. Sample site photograph and ground-truth structured label for an SD-Grounded training case. The label encodes storey, IFC class, and a `spatial_relations` list with predicate, object type, material, and confidence. Crucially, it is derived deterministically from the mined skeleton and wall-region enrichment, rather than from post-hoc annotation.

The curation follows an IFC-first logic: each case begins as a deterministic *skeleton*—a template encoding the target element, its ambiguity pattern, and the retrieval constraints—mined from IFC semantics before any stochastic *skin* (chat, rendered views, floorplans, synthetic site images) is added. This sequence prevents the visual layer from redefining the ground truth after the fact. The following subsections detail the target-pattern mining (4.2.1) and the iterative dataset development (4.2.2).

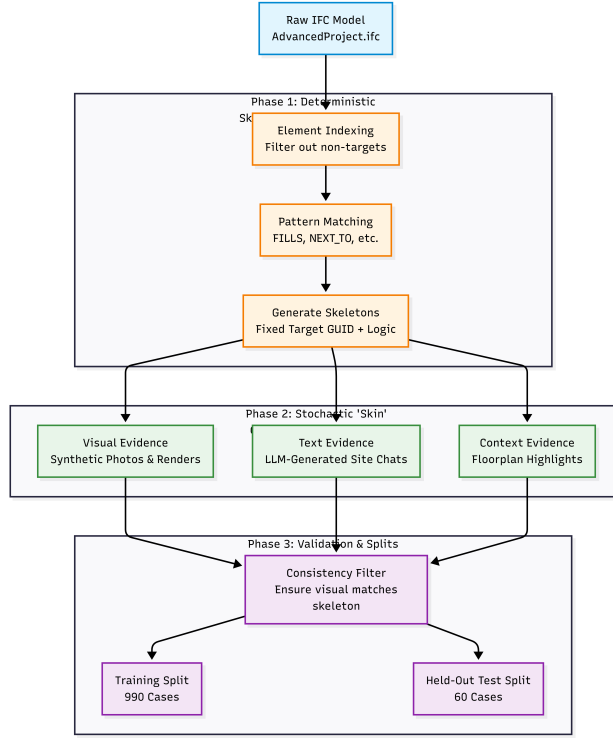


Figure 4.4. Synthetic dataset curation pipeline. Each case is built skeleton-first from IFC semantics before stochastic skin evidence is added, ensuring ground truth is fixed prior to augmentation. The final filter yields 990 training cases (AP only), a 60-case AP held-out [4.2.2.1](#) evaluation set, and a 116-case unified benchmark used as a cross-IFC generalization check.

4.2.1 Target Pattern Mining from IFC

The curation process begins by building a deterministic element index from each IFC model, covering element identity, IFC class, storey, and topological properties. Non-target entities (e.g., abstract spaces, openings, structural subcomponents) are explicitly excluded. A pre-mining check confirms that repeated elements and spatial relationships occur across all three models, demonstrating that the retrieval challenge is not unique to `AdvancedProject.ifc`. Target selection is pattern-based rather than random. The objective is to identify elements whose retrieval requires more than coarse filtering by class or storey. As summarized in Table [Table 4.2](#), this includes repeated-element conditions and topology-bearing relations such as `FILLS`, `ADJACENT_TO`, and `CONTINUOUS`.

The result of this stage is a set of *skeletons*.

Pattern Type	Mining Logic	Dataset Status	Intended Challenge
FILLS	Door/window linked to host wall opening via <code>IfcRelFillsElement</code>	69 KEEP (primary)	Host-element topology reduces otherwise identical candidates
CONNECTS_TO	Wall-to-wall junction derived from shared vertices or proximity in IFC graph	69 KEEP (primary)	Wall-connectivity disambiguation when type and storey are shared
NEXT_TO	Consecutive openings on the same host wall, ordered by position index	97 KEEP (primary)	Left/right ordering of co-planar openings on a common wall
ADJACENT_TO	Spatially neighbouring elements within 1,500 mm centroid distance	62 KEEP (supplementary)	Cross-type proximity; not wall-grounded topology
CONTINUOUS	Multi-storey element whose <code>Top Constraint</code> Pset entry differs from <code>Base Constraint</code> in the Revit-style <code>Constraints</code> property set; extracted from element index without geometry	22 skeletons mined; no skins/cases generated — reserved for future extension	Span-range disambiguation: elements sharing class and base storey are distinguished only by multi-storey reach (base→top constraint pair)

Table 4.2. Target pattern families mined from `AdvancedProject.ifc` for SD-Grounded case construction. `FILLS`, `CONNECTS_TO`, and `NEXT_TO` serve as the primary wall-grounded families.

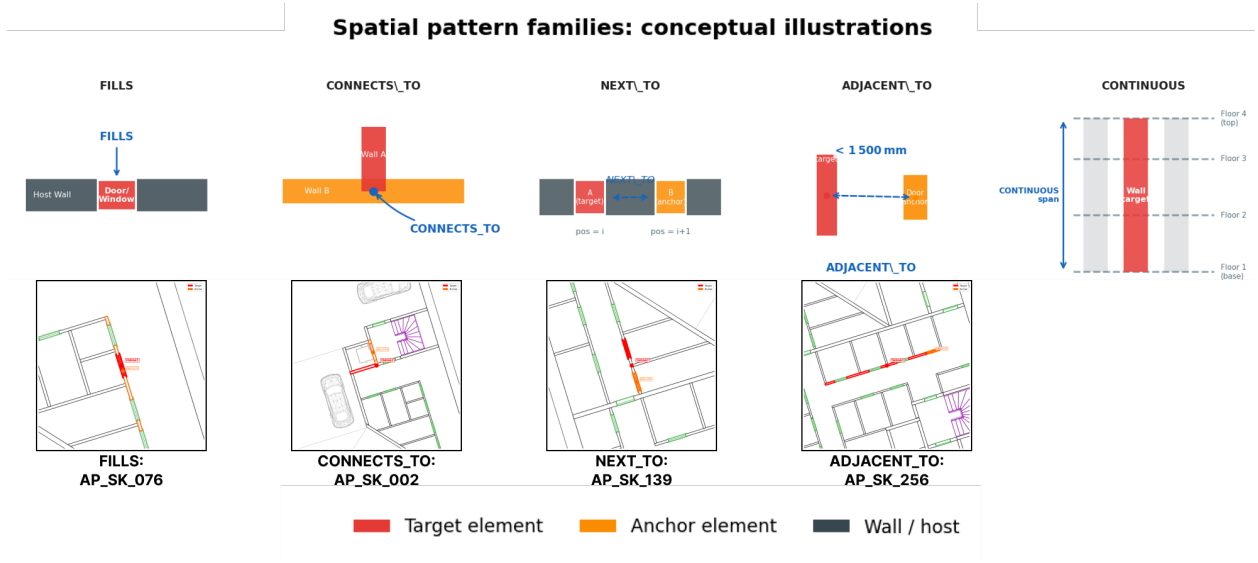


Figure 4.5. Visual sample of target pattern families.

4.2.2 Iterative Development and Final Dataset Composition

To ensure the neural model learns structural reasoning rather than superficial patterns, the dataset underwent four major curation iterations (Table [Table 4.3](#)). The transition from **SD-Baseline** to **SD-Grounded** marks the shift from random element sampling to deterministic, region-grounded supervision.

Iteration	Primary Characteristic	Limitation Revealed
V1	Random IFC element sampling	Too many trivial cases; insufficient retrieval ambiguity
V2	Stratification around repeated element classes	Evidence too simplified; limited multimodal ambiguity
SD-Baseline	Deterministic skeleton mining; multimodal skinning; 1,064 samples (AP, BH, DXA)	Only 37 unique output templates; model learned template shortcuts, not topology
SD-Grounded	Region-grounded supervision; LLM skeleton judging; 990 training samples (AP only)	$\text{CONTINUOUS} = 0$ training cases; storey coverage skewed to Level 1

Table 4.3. Iterative dataset development. Each iteration revealed a distinct limitation that motivated the subsequent design change.

The final composition of both datasets is detailed in Tables [Table 4.4](#) and [Table 4.5](#). Moving forward, the quantitative evaluations in Chapter 5 rely exclusively on the artifacts from **SD-Grounded**. Specifically, the 60-case AP held-out [4.2.2.1](#) set serves as the primary benchmark for evaluating topological retrieval, while the 116-case unified set functions as a secondary check for cross-IFC generalization.

Artifact	Count	Role in Evaluation
Topology skeletons (<code>synth_v0.5</code>)	255	Deterministic templates mined from <code>AdvancedProject.ifc</code>
Training split	1,064	Baseline fine-tuning; AP, BH, DXA
Unified test set	116	End-to-end eval across three IFC models
H2 hard-negative set	568	Symbolic stress test

Table 4.4. SD-Baseline artifact snapshot.

Artifact	Count	Role in Evaluation
AP canonical skeletons (KEEP)	297	LLM-judged templates from <code>AdvancedProject.ifc</code>
Training split	990	237 canonical + 753 augmented; AP only
AP held-out set	60	Primary benchmark; topology-stratified
Unified benchmark	116	Secondary cross-IFC generalization check
Wall-region index	2,277 regions	Region-grounded supervision sidecar

Table 4.5. SD-Grounded artifact snapshot. The primary evaluation in subsequent chapters utilizes the 60-case AP held-out benchmark.

4.2.2.1 Evaluation Dataset

The 60-case AP held-out set from **SD-Grounded** dataset serves as the primary benchmark for evaluating topological retrieval, data flow view at [Figure 4.2](#), provides a stricter evaluation setting, stratified by topology family and relation multiplicity (1-rel, 2-rel, and 3-rel) and organised into six universe families (U1–U6) that capture structurally distinct spatial configurations. As shown in [Figure 4.6](#), the benchmark is dominated by connectivity

singletons, adjacency singletons, paired opening cases, and symmetric triads, rather than labelled deep-hop chains. This distribution follows the realistic prevalence of spatial-relation patterns in the AdvancedProject model and motivates evaluating topology-aware retrieval across multiple relation structures rather than only single-relation cases.

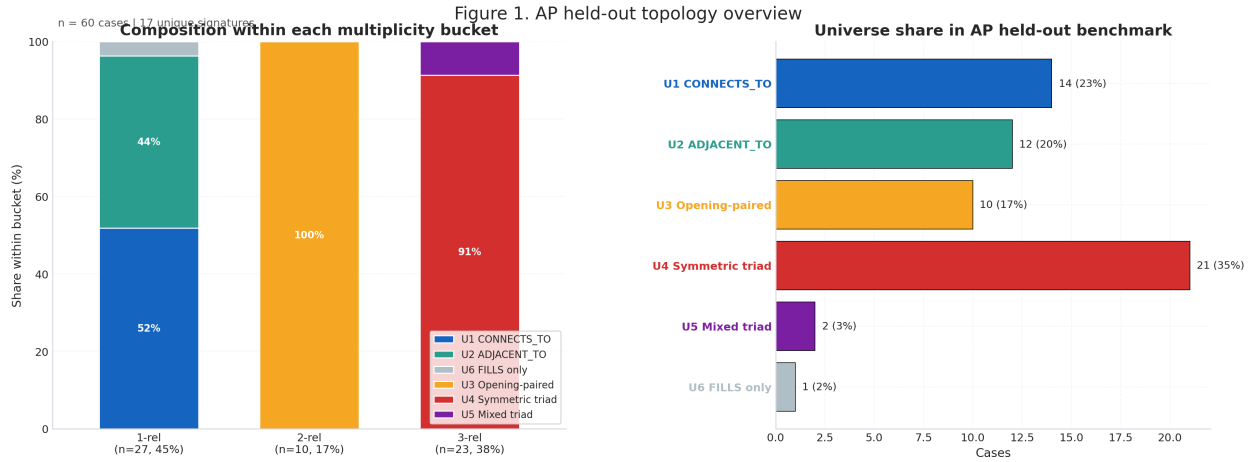


Figure 4.6. AP held-out topology overview. The left panel shows topology-family composition within each relation-multiplicity bucket (1-rel, 2-rel, 3-rel); the right panel summarises the same 60 cases across six universe families (U1–U6). The benchmark is dominated by connectivity singletons, adjacency singletons, paired opening cases, and symmetric triads, motivating topology-aware retrieval evaluation.

4.3 Domain Validity and Limitations

Having established the synthetic curation pipeline in Section 4.2, it is necessary to acknowledge that relying on a semi-synthetic dataset inherently introduces specific boundaries to the evaluation framework.

Furthermore, while the generated multimodal “skin” (LLM-authored chats, synthetic images, floorplans detailed in Section 4.2) maintains internal consistency, it lacks the extreme visual occlusion, lighting variance, and communication errors typical of real-world data.

Crucially, this evaluation assumes the physical site perfectly matches the BIM graph, whereas real PPVC workflows frequently suffer from as-built deviations from the as-designed IFC model. Consequently, while this synthetic sandbox rigorously demonstrates the middleware’s theoretical capacity to address both the gap demonstrate at research questions, multimodal visual grounding (RQ1) and the ontology-driven retrieval (RQ2) identified in Chapter 1.4, bridging this subsequent “Sim-to-Real” domain gap requires future validation on unconstrained physical datasets.

Chapter 5

Experiment and Evaluation Methodology

5.1 Experimental Design Overview

The primary objective of this chapter is to establish a rigorous evaluation framework to diagnose the proposed neuro-symbolic interpreter middleware. In compound AI systems, evaluating only the final retrieval output obscures the root causes of failure. Specifically, end-to-end metrics cannot distinguish whether an error occurred because the neural model hallucinated a spatial constraint or because the symbolic planner failed to execute a valid query against the building graph.

To systematically decouple and measure these layers, the experimental design directly addresses the core research questions and hypothesis established in Chapter 1.4 and 3.3. The methodology follows a logical progression: establishing a baseline floor, determining the theoretical ceiling, evaluating the actual system, and conducting deep diagnostics. This progression is structured into four experimental groups, mapped to their respective Research Questions (RQs) in Table 5.1 and would further explain below at 5.3

Table 5.1. Mapping of Experimental Groups to Core Research Questions

Group	Evaluation Phase	Core Objective	Addressed RQ
Group 1	System Baselines	Establish the performance “floor” using pure LLM agents and non-topological methods.	RQ2
Group 2	The Oracle Ceiling	Determine the theoretical limit of the symbolic retrieval layer under perfect neural extraction.	RQ2
Group 3	Dual-Track Evaluation	Test multimodal spatial grounding and end-to-end GUID retrieval.	RQ1
Group 4	Post-Hoc Diagnostics	Analyze system robustness, true graph ambiguity, and hallucination mitigation.	RQ1 & 2

5.2 Evaluation Metrics

Before evaluating the experimental groups, standard information retrieval metrics are defined to ensure reproducible comparisons. Let N represent the total number of query cases, GT_i represent the ground-truth Global Unique Identifier (GUID) for the i -th case, and $\mathcal{P}_i$ represent the pool of candidate GUIDs returned by the symbolic planner.

5.2.1 Neural Extraction Quality (Intermediate Diagnostics)

Extraction metrics isolate the contribution of the learned Vision-Language Model (VLM) front-end by evaluating whether it produces valid constraints prior to symbolic retrieval. This is assessed through per-field accuracy and overall training performance.

5.2.2 End-to-End Retrieval Metrics (Intermediate Diagnostics)

Retrieval metrics quantify whether the interpreter successfully grounds multimodal evidence to the correct IFC entity using the symbolic graph. Let n represent the total number of query cases, GT_i represent the ground-truth element, and Pool_i represent the candidate pool returned by the planner.

Ground-Truth in Pool (GT-in-Pool): This metric establishes a pipeline health check by evaluating whether the correct element survives symbolic filtering. It serves as a necessary precondition for subsequent ranking metrics.

$$\text{GT-in-Pool} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}[\text{GT}_i \in \text{Pool}_i] \quad (5.1)$$

Top- k Hit Rate and MRR@10: The Top- k Hit Rate measures strict precision by recording the fraction of cases where the ground truth appears within the first k results. Mean Reciprocal Rank (MRR@10) further captures early-rank quality by rewarding systems that place the target closer to the first rank position. Let rank_i be the position of GT_i in Pool_i :

$$\text{Top-}k = \frac{1}{n} \sum_{i=1}^n \mathbf{1}[\text{rank}_i \leq k] \quad (5.2)$$

$$\text{MRR@10} = \frac{1}{n} \sum_i \frac{1}{\text{rank}_i} = \frac{1}{n} \sum_i \frac{1}{\text{rank}_i}, \text{ where } \frac{1}{\text{rank}_i} = 0 \text{ if } \text{rank}_i > 10 \text{ or } \text{GT}_i \notin \text{Pool}_i \quad (5.3)$$

Recall-Weighted Reduction (RWR): Search Space Reduction (SSR) can be misleading if the correct element is discarded. RWR ensures the system is not rewarded for "aggressive" filtering that loses the target, assigning zero credit when the ground truth is lost. Where N is the total count of indexed elements in the IFC model:

$$\text{RWR} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}[\text{GT}_i \in \text{Pool}_i] \cdot \left(1 - \frac{|\text{Pool}_i|}{N}\right) \quad (5.4)$$

Metric	Definition	Interpretive Role
GT-in-Pool	Fraction of cases where the ground-truth element survives into the candidate pool	Pipeline health check; a precondition for valid Top- k evaluation
Top- k Hit Rate	Fraction of cases where GT appears within the top k ranked results ($k \in \{1, 10\}$)	Primary retrieval quality signal; Top-1 measures strict precision, Top-10 measures practical utility
MRR@10	Mean reciprocal rank of the first GT hit within top 10	Captures early-rank quality; more sensitive to rank position than Top- k
Recall-Weighted Reduction (RWR)	Relative reduction in candidate pool size versus the full type-matched baseline	Measures how much the symbolic filter narrows the search space
Avg Pool Size	Mean number of candidates remaining after retrieval	Absolute measure of disambiguation degree; complements SSR

Table 5.2. End-to-end retrieval metrics.

Metric	Definition	Interpretive Role
VLM Inference Latency	Wall-clock time for the multimodal extractor to produce structured constraints	Dominant cost in the pipeline; determines interactive feasibility
Cypher Query Latency	Wall-clock time for Neo4j to execute the symbolic retrieval query	Secondary cost; typically negligible relative to VLM inference

Table 5.3. System-level metrics.

5.3 Experiments Design

5.3.1 Group 1: End-to-End System Baselines

Before evaluating the fine-tuned multimodal models, it is necessary to establish the baseline performance of standard retrieval paradigms. This group establishes the operational baseline of the evaluation.

Agentic Pipeline Baseline This baseline evaluates whether a constrained neuro-symbolic pipeline provides a more reliable interpretation workflow than a free-form agentic approach. The comparison is framed between two architectures: an agent-driven pipeline where the model retains open-ended control over tool invocation, and a structured pipeline where constraint extraction, query planning, and retrieval are explicitly separated. The purpose is to determine whether reducing interpretive freedom improves the reliability of site-to-model alignment in high-ambiguity AEC contexts.

Attribute-Only Retrieval (P0 Strategy) This baseline bypasses multimodal spatial reasoning entirely. It queries the IFC graph using only standard semantic attributes (e.g., IFC class, storey) without any topological constraints. This experiment empirically quantifies the baseline attribute entropy of the PPVC dataset, proving whether structural topology is an absolute necessity for disambiguation.

Foundational VLM Baseline Representing the standard generative AI approach, this baseline utilizes a state-of-the-art Vision-Language Model (e.g., Gemini) in a prompt-only capacity, without domain-specific LoRA fine-tuning. It tests whether a foundational model can bridge visual site evidence and IFC topologies purely through in-context learning.

5.3.2 Group 2: The Oracle Ceiling and Symbolic Limits

In a neuro-symbolic architecture, end-to-end failure can originate from either a neural hallucination or a symbolic planner limitation. To address RQ2, this group isolates the symbolic layer by establishing the **Oracle Ceiling**.

Ceiling Performance With Perfect Extraction This experiment asks: “*What is the theoretical maximum performance if neural perception is flawless?*” In problem environments characterized by a high density of identical elements [1.3.2](#), consistently achieving a perfect Top-1 retrieval may be a mathematically impossible target. Therefore, an Oracle is evaluated to establish the true performance ceiling and accurately identify operational bottlenecks. The probabilistic VLM is bypassed entirely, and the symbolic planner is fed perfect, ground-truth extractions of topology, direction, and subtype directly from the dataset.

Query Planner Strategy Ablation Given the domain challenges of highly identical elements and limited visual attributes, this experiment evaluates a spectrum of symbolic retrieval strategies to identify the optimal approach for disambiguation. The Oracle is tested across a progressive ladder of strictness, ranging from simple attribute-based queries (L0)

to complex, multi-hop topological fingerprints (L4), alongside corresponding query planner priority levels (P0-P8).

5.3.3 Group 3: Dual-Track Evaluation of Vision-Language Models

Having established the theoretical ceiling (Group 2), this group constitutes the primary evaluation of the candidate models to address RQ1. To rigorously decouple neural extraction errors from downstream database limitations, the pipeline is evaluated across two parallel tracks.

Track A: Neural Extraction Quality (Intermediate Diagnostics) Track A isolates the Vision-Language Model (VLM). It measures how accurately the neural layer translates multimodal site evidence into strict, ontology-compliant constraints (e.g., predicate, direction, subtype). By isolating this initial step, the evaluation specifically captures neural hallucinations and extraction fidelity without conflating them with symbolic execution logic (evaluated via metrics in Section 5.2).

Track B: End-to-End Retrieval Accuracy (System Validation) Track B evaluates the neuro-symbolic pipeline end-to-end. It compares practical retrieval performance across successive architectural iterations, ranging from zero-shot baselines and attribute-only extractors to fully augmented topological models and Graph-RAG interventions. As the ultimate test of real-world applicability, Track B measures how effectively the combined system grounds visual evidence to exact IFC element GUIDs (evaluated via metrics in Section 5.2).

Candidate Model Configurations

Rather than evaluating a single monolithic model, this dual-track methodology is applied across a progressive series of fine-tuned extractors. The complete roster of evaluated models—detailing their training augmentation, parameter configurations, and architectural evolution—is documented in Chapter 6 (specifically Section 6.2.2 and Table 6.2).

5.3.4 Group 4: Ablation and Post-Hoc Analyses

The fourth experiment group tests robustness, failure modes, and generalisation. Whereas the earlier groups establish the main architectural and model-level effects, these analyses ask how strongly those effects depend on particular modalities, input cues, or IFC sources, and whether the fine-tuned models exhibit unintended learning shortcuts. Group 4 comprises seven branches.

Modality Ablation. To determine which input modalities contribute most effectively to accurate constraint extraction, the same cases are evaluated across five evidence configurations: floorplan-only (FP), site-photo-only (SITE), text plus site photo (MB), text plus floorplan (MC), and all modalities (MA). Preliminary evaluations indicate that site photos yield a higher Ground Truth (GT)-in-Pool rate than floorplans (by approximately 10 percentage points), and aggregating all modalities does not strictly guarantee performance gains over the optimal single modality (detailed in Chapter 7).

Model Diagnostics. The evaluation framework employs decomposed diagnostics to isolate failure modes and guide targeted pipeline improvements:

- **Shortcut Learning Analysis:** This assesses whether fine-tuned models genuinely condition their spatial-relation outputs on visual evidence, rather than relying on textual priors. By comparing extraction variances across modality pairs (e.g., FP vs. MC, MA vs. FP), this analysis quantifies the actual impact of image content on predicted spatial relations.
- **Per-Field and Multi-Hop Accuracy:** This measures extraction fidelity at a granular level, specifically quantifying per-hop accuracy and the rate of hallucinated topological hops. This evaluation tests whether upstream extraction improvements reliably translate into downstream retrieval quality, demonstrating the necessity of joint metric evaluation.

Ultimately, this decomposable diagnostic approach exposes the operational capabilities and limitations of the model. It identifies the most effective input combinations, enables highly targeted debugging, and provides actionable guidance for practical deployment.

Graph-RAG Reranker. To assess the efficacy of graph-based reranking, retrieval performance on the AP held-out benchmark (Section 4.2.2.1) is evaluated under two primary conditions: with the Graph-RAG module applied, and without it. The objective is to determine whether secondary LLM-driven reranking provides additive value depending on the quality of the initial retrieval pool.

Chapter 6

Prototype Implementation

This chapter details the technical implementation of the neuro-symbolic retrieval middleware. It translates the conceptual frameworks established in Chapters 3 into a deployable software architecture, demonstrating the functional prototype that serves as the foundation for all subsequent experiments and evaluations 5.

6.1 System Architecture Overview

The implemented prototype is a compound AI system designed to address the ambiguity-prone workflows of high-rise building construction (Section 1.4.2). Rather than relying on a single end-to-end black-box model, the system employs a **Hierarchical Neuro-Symbolic Architecture** (Figure 6.1) as mentioned in 3.3.1. Neural components (Vision-Language Models) provide multimodal perception, while symbolic structures (IFC graphs and JSON schemas) enforce geometric and regulatory logic.

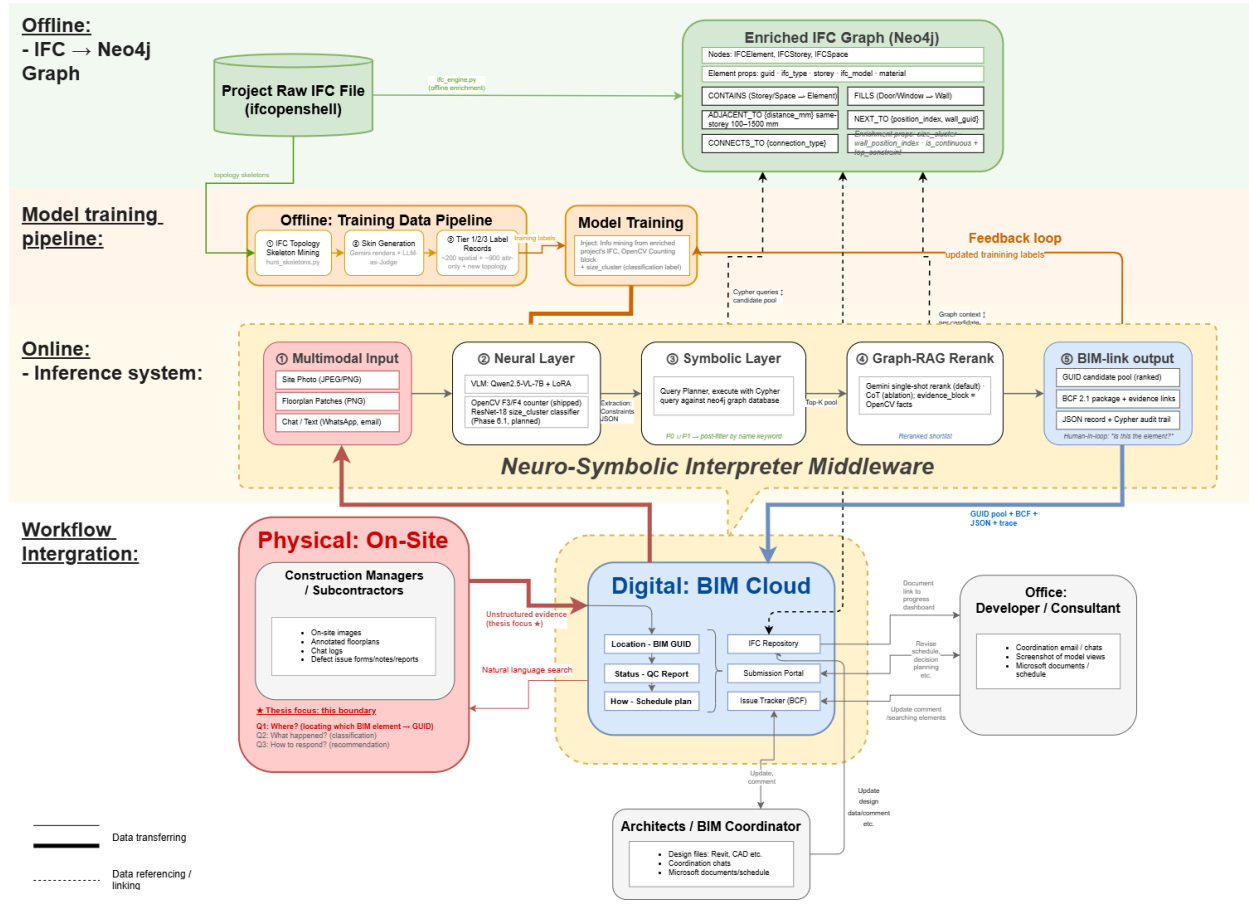


Figure 6.1. Neuro-symbolic pipeline detail. Multimodal inputs (site photo, floorplan, chat, optional 4D context) are processed by the neural layer through one of four extractor variants. The resulting Constraints JSON feeds the symbolic layer, which executes a priority-ordered query cascade (P0a/P0b, P1–P2, P4–P6, P8) against the Neo4j IFC graph. The validation and handoff stage packages results into GUID pools, JSON records, and BCF packages for downstream CDE integration.

6.1.1 Runtime Inputs, Reference Resources, and Outputs

To maintain this architectural boundary, the pipeline strictly distinguishes between dynamic site evidence and static ground-truth resources. At runtime, the ingestion layer aggregates multimodal inputs into a unified prompt payload, while the IFC model serves as the immutable geometric reference.

• Runtime Inputs:

- *Site Evidence*: Chat excerpts, issue descriptions, site photos, rendered floorplans, and 4D context (e.g., task status or schedule metadata).

- **Reference Resources:**

- *IFC Model*: The authoritative geometric and topological source utilized for graph retrieval and spatial grounding.
- *JSON Schema*: A validation template informed by CORENET X requirements, used to structure the downstream record once the target element is retrieved.

- **System Outputs:**

- *Structured Constraints*: The machine-readable intermediate JSON representation extracted by the neural layer.
- *Traceable BIM Reference*: The retrieved candidate pool and the ultimate target IFC element GUID.
- *Validated Record*: A finalized, structured output validated against the domain schema.
- *Handoff Artifact*: A BCF 2.1 package compiled for seamless downstream Common Data Environment (CDE) integration.

- **Evaluation Artifacts:**

- *Execution Traces*: JSONL or CSV logs captured at the boundary interfaces, utilized for experiment analysis, debugging, and diagnostic metric reporting.

6.1.2 Core Modules

The end-to-end pipeline is implemented across five primary codebase modules:

- **Neuro Extractor (`src/neurosym/`):** Prompt- and LoRA-based Vision-Language Models that parse multimodal evidence into typed JSON constraints, supported by secondary modules like OpenCV and ResNet for visual heuristic bounding.
- **IFC Engine (`src/ifc_engine.py`):** The domain grounding layer responsible for IFC parsing, geometric normalisation, graph topology enrichment, and storey-aware indexing.
- **Symbolic Query Planner & Retrieval Backend (`src/neurosym/constraints_to_query.py`, `src/neurosym/retrieval_backend.py`):** Deterministic modules that compile JSON constraints into explicit Cypher query plans and execute retrieval against the Neo4j database, including optional Graph-RAG reranking mechanisms.

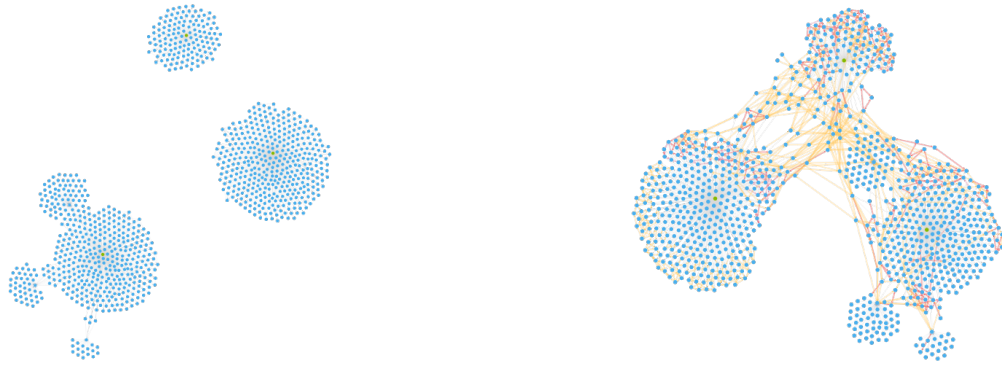
- **Validation and Handoff** (`src/rq2_schema/`, `src/handoff/`): Deterministic components that enforce JSON Schema checks, validate domain compliance, and package the output artifacts.
- **Data Mining & Feedback Pipeline**: An offline infrastructure module that generates paired training datasets from the enriched IFC graph and manages the continuous feedback loop for subsequent model refinement.

6.1.3 IFC Engine: From Raw IFC to Enriched IFC Graph

Implemented primarily in `src/ifc_engine.py`, the IFC engine functions not as a passive parser, but as an active transformation layer that converts raw BIM data into a normalized, retrieval-oriented knowledge graph [26, 45]. Beyond extracting basic architectural entities and containment hierarchies, the engine constructs canonical registries for storeys, spaces, and materials. Because raw IFC exports suffer from severe naming inconsistencies (e.g., `Level 1` versus `1 - First Floor`), the engine applies LLM-assisted parsing with deterministic fallbacks to resolve these brittle strings into stable query targets [21, 43]. This normalization is critical for cross-project generalisation, absorbing the naming variance that would otherwise disrupt cross-model retrieval (as evaluated in Chapter 7).

Beyond attribute normalization, the engine performs topological graph enrichment and schema compression to overcome the sparse connectivity of native IFC links, the engine computationally derives physically meaningful spatial relations, that are essential for disambiguating repetitive architectural layouts [26, 32]. [Table 6.1](#) and [Table 4.2](#) details the geometric and semantic operations driving this enrichment.

The structural impact of this enrichment is visualized in Neo4j. As illustrated in [Figures 6.2](#) and [Figure 6.3](#), the algorithmic enrichment transforms the base hierarchical skeleton into a densely connected retrieval graph.



(a) Global base graph overview

(b) Global enriched graph overview

Figure 6.2. Global comparison between the base IFC graph and the enriched IFC graph. The base view is dominated by containment and fill relations, whereas the enriched view becomes more densely connected through retrieval-oriented topology edges such as `NEXT_TO`, `CONNECTS_TO`, and `ADJACENT_TO`.

Table 6.1. Normalization operations used by the IFC engine in the implemented system.

Item	Source	Method / formula	Retrieval role	Code location
Storey registry	Naming normalization	LLM parse with regex fallback to canonical floor number and level label.	Robust level grounding	<code>src/ifc_engine.py</code>
Space registry	Naming normalization	LLM parse with fallback rules for room type and unit identifiers.	Finer spatial grounding	<code>src/ifc_engine.py</code>
Material field	IFC material association	Flatten <code>IfcRelAssociatesMaterial</code> into canonical material strings.	Post-filtering and semantic refinement	<code>src/ifc_engine.py</code>

At a local scale, the same enrichment can be inspected around a representative anchor element show at [Figure 6.3](#). This illustrative: it makes explicit that the local graph used for retrieval is not limited to storey containment and fill relations, but also exposes lateral links such as `NEXT_TO`, `CONNECTS_TO`, and `ADJACENT_TO` when they are present.


```
[
  {
    "role": "system",
    "content": "<lora_system_g9      see Appendix B>"
  },
  {
    "role": "user",
    "content": [
      {"type": "image", "image": "file:///.../AP_SK_294_site.png"},
      {"type": "image", "image": "file:///.../AP_SK_294_scale_M.png"},
      {}
    ],
    {
      "type": "text",
      "text": "[4D Task Status] Window Installation\n[Project Phase] Facade Works\n[Chat Log]\n supervisor: On Level 1, check the window beside the door at this location.\n[OpenCV Counting] 2nd of 3 openings on the same wall\n[Query] identify the BIM element\nExtract the search constraints as JSON."
    }
  ]
}
```

6.2.1 Textual Pre-Clustering for Dimension Constraints

To bypass the limitation of VLMs fail on absolute dimensions measuring from visual evidence such as floorplans, the first stage of the hybrid pipeline applies a lightweight KMeans clustering algorithm to the textual context. This pre-clustering classifies target elements into relative dimensional buckets, outputting a deterministic `size_cluster` constraint. This approach provides the downstream planner with reliable, dimensionally aware filters without demanding impossible visual precision from the neural network.

6.2.2 Model Deployment and Parameter-Efficient Fine-Tuning (PEFT)

The core semantic parsing relies on the Qwen2.5-VL-7B-Instruct backbone, deployed using Low-Rank Adaptation (LoRA) in 4-bit quantization via Unsloth on Modal A100 GPUs [25, 36]. The neural extractor evolved through successive fine-tuning generations to sys-

tematically improve spatial predicate support, multi-hop relation extraction, and floorplan-centered grounding.

Table 6.2 summarizes the evolutionary milestones of the neural extractor. All training iterations utilize the Qwen2.5-VL-7B-Instruct 4-bit backbone deployed via Modal A100 GPUs. The model progression begins with a baseline restricted to attribute-only extraction, systematically evolving to support multi-hop spatial predicates and cross-model floorplan grounding. Later iterations explore hyperparameter sensitivities and introduce increasingly rich spatial fingerprints, incorporating relative direction and physical dimensions. The culminating iteration specifically trains the VLM to act purely as a semantic parser—extracting foundational attributes (e.g., `storey_name`, `ifc_class`) and topological `spatial_relations`—while intentionally delegating precise ordinal counting to parallel deterministic modules.

Table 6.2. Consolidated master roster of evaluated extractors. The progression tracks the architectural shift from zero-shot prompting to hybrid, topology-rich spatial parsing.

ID	Training Data	Settings	Primary Purpose / Key Architecture
Zero-Shot & Foundational Baselines			
Gemini 2.5	None (Zero-shot)	Prompt	Standard generative multimodal reference.
Gemini AP	None (Zero-shot)	Schema prompt	Schema-corrected, structured zero-shot reference.
Phase 1: Topological Proof-of-Concept (Early LoRA Series)			
LoRA_2	AP-only	$r = 16$	Baseline restricted strictly to attribute-only extraction.
LoRA_4	560 train / 75 test	$r = 16$, $lr\ 2 \times 10^{-4}$	Introduced 4 predicates and two-hop spatial retrieval capability.
LoRA_5	1,064 train / 92 test	$r = 32$, $lr\ 2 \times 10^{-4}$	Added NEXT_T0; trained across 3 IFC models with floorplan emphasis.
Phase 2: Canonical Fine-Tuning, Augmentation, & Hybrid Vision (G-Series)			
G0	237 canonical (AP)	$r = 16$, $lr\ 1 \times 10^{-4}$	Region-grounded supervision without data augmentation.
G1-G2	990 full augmented	$r = 16$, lr variable	Explored augmentation effects and learning rate sensitivity.
G3	990 full augmented	$r = 32$, $lr\ 1 \times 10^{-4}$	Optimized for recall.
G4	990 full augmented	$r = 32$, $lr\ 1 \times 10^{-4}$	Optimized for precision; highest Hop-1 extractor (Track A).
G6	990 full augmented	$r = 32$, $lr\ 1 \times 10^{-4}$	Underfit control (Checkpoint-20) for failure referencing.
G7	990+ position labels	$r = 32$, $lr\ 1 \times 10^{-4}$	Richer supervision (position_context , direction); highest MRR@10.
G8	Enriched labels	$r = 32$, $lr\ 1 \times 10^{-4}$	Integrated absolute physical dimension targets.
G9	Hybrid augmented	$r = 32$, $lr\ 1 \times 10^{-4}$	Culminating hybrid: VLM semantic parsing explicitly paired with OpenCV and ResNet.

6.2.3 Deterministic Visual Heuristics

To resolve visual ordinal counting as a hard empirical ceiling for current VLMs, the pipeline integrates OpenCV-based contour analysis as a deterministic visual heuristic. OpenCV scans the floorplan regions identified by the VLM to explicitly isolate and count repeating geometric elements (e.g., identical window bays). This module outputs a mathematically verifiable

`position_context` (e.g., "3rd of 17 openings"), structurally augmenting the VLM’s probabilistic spatial triplet with a precise ordinal count.

In parallel, a ResNet classifier is introduced to predict element size categories from local image patches, providing a complementary discriminative signal where geometric counting alone is insufficient.

We note two limitations in this prototype implementation. The goal at this stage is solely to demonstrate that OpenCV and ResNet can provide reliable counting and classification signals to the symbolic layer; accordingly, (i) the wall section used for OpenCV counting is hard-coded, and (ii) the `world_xy` coordinates consumed by ResNet are precomputed rather than inferred end-to-end. Full integration with upstream region proposal is left to future work.

Table 6.3. OpenCV floorplan counter parameters.

Parameter	Value
Gaussian blur	$5 \times 5, \sigma = 0$
Canny thresholds (wall search)	60 / 160
Hough ρ / θ	1 px / $\pi/180$
Hough vote threshold	20
Min line length / max gap	14px / 18px
HSV saturation mask	$S \geq 45$
Morphological open kernel	3×3
Min component area / aspect	10px ² / 1.3
Confidence override gate	0.80

Table 6.4. ResNet-18 size-band classifier configuration.

Parameter	Value
Backbone / weights	ResNet-18, ImageNet1K-V1
Head	<code>Linear(512, 6)</code>
Input	192×192 RGB, ImageNet norm.
Augmentation	rot {0, 90, 180, 270}, ± 8 px, jitter (0.15, 0.10)
Loss	weighted cross-entropy (inverse freq.)
Optimiser / LR / WD	AdamW / 3×10^{-4} / 1×10^{-4}
Schedule	CosineAnnealing, $T_{\max} = 30$
Epochs / batch	30 / 16
Checkpoint	min(val_loss)
Split (train/val/test)	155 / 17 / 23
Test accuracy	82.6%

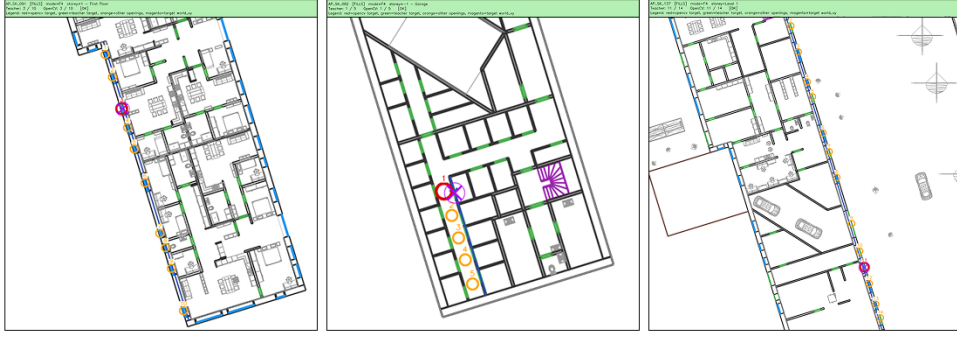


Figure 6.4. OpenCV-based contouring visualization sample. Red circle is the target and yellow circle is the valid counts, which able to extract `position_context` such as "3rd of 17 openings".

6.2.4 Constrained JSON Decoding and Schema Enforcement

The final stage of the neural layer is a deterministic augmentation pipeline that fuses the outputs of three extraction streams—the `size_cluster`, the OpenCV `position_context`, and the VLM’s semantic extractions (including core attributes like `storey_name` and topological `spatial_relations`)—into a single, unified JSON payload. The inference module programmatically enforces this schema, guaranteeing a 100% parse rate for the downstream symbolic planner.

Table [Table 6.5](#) details the top-level `Constraints` schema, while Table [Table 6.6](#) outlines the nested topology fields. Notably, the deterministic `position_context` is passed to the symbolic planner, where it is later mapped to a `wall_position_index` for precise structural querying. Listing [Listing 6.2](#) shows the normalized JSON object produced for the running case AP_SK_294 after the three extraction streams have been merged and decoded.

Table 6.5. Top-level fields in the **Constraints** contract (G9 + Phase 6.1).

Field	Type	Role	Example
storey_name	optional string	Floor cue for early pool reduction.	"6 - Sixth Floor"
ifc_class	optional string	Target IFC entity class.	"IfcWindow"
space_name	optional string	Room / zone cue.	"Living Room"
target_name_-keyword	optional string	Rerank-only descriptor; bridges human label $\leftrightarrow$ IFC name.	"floor-to-ceiling window"
position_context	optional string	OpenCV-validated slot-aware anchor.	"3rd of 17 openings"
position_-context_-confidence	optional float	Counter confidence (≥ 0.8 triggers override).	0.92
position_-context_source	optional string	Provenance tag.	"opencv_f4"
size_cluster	optional string	Rerank-only soft signal; dimension-anchor taxonomy.	"window_M_-1480x1380"
size_band	optional string	ResNet-predicted 6-way band; <i>retrieval filter</i> .	"window_M"
size_band_-confidence	optional float	Classifier softmax; gates hard filter (≥ 0.6).	0.83
size_band_source	optional string	Origin tag.	"resnet_-oracle_-centroid"
spatial_-relations	list of SpatialTriplets	Structured topology for P0 retrieval.	Table 6.6
confidence	float	Overall extraction gate.	0.85
source	string	Extractor route: prompt/lora/prompt_-failed.	"lora"

Table 6.6. Per-triplet fields in the SpatialTriplet contract.

Field	Type	Role	Example
subject_type	string	Target IFC class.	"IfcWindow"
predicate	enum (8)	FILLS, CONTINUOUS, ADJACENT_TO, NEXT_TO, CONNECTS_TO, ON_TOP_OF, PERPENDICULAR_TO, PARALLEL_TO.	"FILLS"
object_type	string	Reference element class.	"IfcWall"
object_subtype	optional string	Subtype fingerprint hint.	"IfcWallStandardCase"
direction	optional enum	left or right along wall axis.	"left"
object_material	optional string	Material filter on reference.	"Concrete"
confidence	float	Per-triplet VLM confidence $\in [0, 1]$.	0.78
connection_degree	optional int	For CONNECTS_TO: 1=end, 2=corner, 3+=junction.	2
distance_mm	optional float	For ADJACENT_TO: centroid distance.	1240.0

Listing 6.2: Extracted Constraints payload for case AP_SK_294.

```
{
  "storey_name": "1",
  "ifc_class": "IfcWindow",
  "space_name": null,
  "target_name_keyword": "floor-to-ceiling window",
  "position_context": "2nd of 3 openings on the same wall",
  "target_width_mm": 1894.0,
  "target_height_mm": 2094.0,
  "spatial_relations": [
    {"predicate": "NEXT_TO", "object_type": "IfcDoor",
      "direction": "right", "object_subtype": "fire-rated door",
      "confidence": 1.0},
    {"predicate": "FILLS", "object_type": "IfcWallStandardCase",
      "object_material": "Brick, Engineering",
      "host_name": "Basic Wall:MockUp Exterior Ground Floor",
      "confidence": 1.0},
    {"predicate": "NEXT_TO", "object_type": "IfcWindow",
```

```

    "direction": "left", "object_subtype": "floor-to-ceiling window
    ",
    "confidence": 1.0}
]
}

```

6.3 The Symbolic Layer: Planner and Retrieval Logic

The symbolic layer acts as the deterministic engine of the system, grounding the augmented JSON constraints extracted by the neural layer against the enriched IFC knowledge graph. This stage enforces engineering truth: rather than probabilistically generating new meaning, it systematically narrows the search space, validates candidate elements, and preserves explicit traceability to authoritative IFC entities.

6.3.1 Query Planner: Constraint-to-Query Compilation

Implemented primarily in `src/neurosym/constraints_to_query.py`, the query planning layer translates the extracted `Constraints` payload into an explicit Cypher retrieval plan. To guarantee deterministic execution, the planner eschews open-ended graph query generation in favour of a strict, priority-ordered cascade([Table 6.7](#)).

This design explicitly separates neural uncertainty from symbolic execution, rendering the retrieval logic fully auditable. The neuro-symbolic path is dominated by Priority 0 (P0) spatial grounding. At this level, the planner leverages the pipeline’s hybrid extraction data, dynamically incorporating highly selective topological edges (`spatial_relations`), OpenCV-validated ordinal positions (`wall_position_index`), and KMeans-derived dimensional constraints (`size_cluster`). If spatial cues are absent or extraction confidence is critically low, the planner degrades gracefully through attribute-based fallback strategies (P4–P6), ensuring the pipeline never fails catastrophically on ambiguous inputs.

Table 6.7. Priority-ordered query cascade implemented in `constraints_to_query.py`. All rules whose required fields are non-empty emit a parallel query plan; the retrieval backend fuses their pools at execution time (Section 6.3.2). Priority 0 splits into two sub-rules because the `CONTINUOUS` predicate is materialized as a node property (`is_continuous`, `top_constraint`) rather than an edge, and therefore requires a different Cypher shape.

Level	Strategy	Main evidence	Role in the cascade
P0a	spatial_triplet (edge)	<code>spatial_relations</code> + <code>ifc_class</code> , predi- cate $\neq$ <code>CONTINUOUS</code>	Edge-traversal grounding via <code>FILLS</code> , <code>NEXT_T0</code> , <code>ADJACENT_T0</code> , etc.
P0b	continuous_ span (prop- erty)	<code>spatial_relations</code> + <code>ifc_class</code> , predi- cate = <code>CONTINUOUS</code>	Property-based filter on <code>is_continuous</code> and <code>top_constraint</code> ; no edge traversal. Used for walls and columns that span multiple storeys.
P1	space+type	<code>space_name</code> + <code>ifc_class</code>	Room-bounded retrieval when a named space is available.
P2	name.keyword	<code>target_name_-</code> <code>keyword</code>	Equipment or subtype retrieval by explicit ID or name fragment.
P4	storey+type	<code>storey_name</code> + <code>ifc_class</code>	Attribute baseline; also used as the P0 safety-net tier under the default union strategy (Section 6.3.2).
P5	storey_only	<code>storey_name</code>	Broader floor-level narrowing.
P6	type_only	<code>ifc_class</code>	Global type filter across all storeys.
P8	fallback	none	Escalation candidate returning a capped broad pool (100 elements).

Foundational attributes (such as `storey_name` and `ifc_class`) compile to direct string-matching filters, while topological predicates (`FILLS`, `NEXT_T0`) and OpenCV-derived heuristics (`direction`, `wall_position_index`) compile to explicit multi-hop graph traversals. Figure 6.5 visualizes the target subgraph pattern for a canonical window-beside-door query.

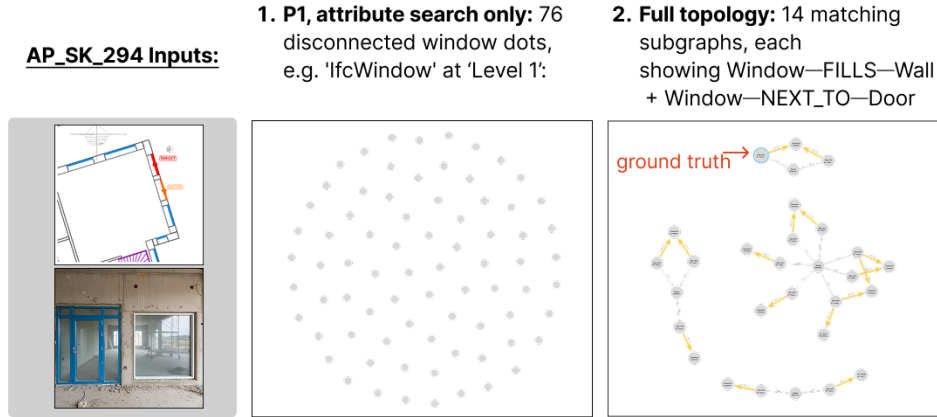


Figure 6.5. Visual representation of the compiled Cypher query executed against the enriched Neo4j graph. The target `IfcWindow` is deterministically grounded via its multi-hop topology: filling a brick `IfcWallStandardCase` while positioned sequentially to the left of an `IfcDoor` along the same wall axis.

Applied to the running case `AP_SK_294`, the same translation rules yield the concrete Priority 0a Cypher shown in Listing [Listing 6.3](#). The three extracted `spatial_relations` from Listing [Listing 6.2](#) compile into a single multi-anchor pattern: one `FILLS` host edge anchoring the target to a brick `IfcWallStandardCase`, and two `NEXT_TO` edges binding the window to a co-wall `IfcDoor` and a sibling `IfcWindow` via the shared `wall_guid` join key. Because two or more triplets are present, the planner routes the plan through the multi-anchor Cypher shape (triggered by `_requires_multi_chain`) rather than the simpler single-hop path.

Listing 6.3: Compiled Priority 0 Cypher for case `AP_SK_294`. Parameters (`$storey_list`, `$fills_obj_0`, ...) are bound from the extracted JSON at execution time.

```
MATCH (target:IFCElement)
MATCH (target)-[:FILLS]->(fi0:IFCElement)
MATCH (target)-[nt_r0:NEXT_TO]->(nt_nb0:IFCElement)
MATCH (target)-[nt_r1:NEXT_TO]->(nt_nb1:IFCElement)
WHERE (target.ifc_type = $subject_type
      OR target.ifc_type STARTS WITH $subject_type)
AND target.ifc_model = $model
AND (size($storey_list) = 0
     OR ANY(s IN $storey_list
            WHERE toLower(target.storey) CONTAINS s))
AND (fi0.ifc_type = $fills_obj_0
     OR fi0.ifc_type STARTS WITH $fills_obj_0)
AND ($fills_mat_0 = ''
     OR toLower(fi0.material) CONTAINS toLower($fills_mat_0))
AND (nt_nb0.ifc_type = $nt_obj_0
```

```

        OR nt_nb0.ifc_type STARTS WITH $nt_obj_0)
AND nt_r0.wall_guid = fi0.guid
AND (nt_nb1.ifc_type = $nt_obj_1
    OR nt_nb1.ifc_type STARTS WITH $nt_obj_1)
AND nt_r1.wall_guid = fi0.guid
AND nt_nb0 <> nt_nb1
RETURN DISTINCT target.guid AS guid, target.storey AS storey,
    target.wall_position_index AS pos
ORDER BY pos;

```

Three design choices visible in this listing deserve attention. First, the IFC subtype-aware clause `STARTS WITH` ensures that the extracted `IfcWallStandardCase` host also matches any future `IfcWall` subtype without planner modification. Second, the `wall_guid` equality join between `nt_r0` and `nt_r1` enforces the “same wall” constraint implicit in the natural-language query, disallowing matches that span adjacent walls. Third, the `nt_nb0 <> nt_nb1` guard blocks degenerate self-matches that would otherwise occur when the door and sibling-window references resolve to the same node.

6.3.2 Retrieval Execution: From Neural Constraints to Final GUID

The retrieval backend (`src/neurosym/retrieval_backend.py`) executes the compiled plans against the Neo4j graph. The Group 2 oracle experiment (Chapter 7) showed that strict multi-hop intersections over-prune the target whenever neural extraction is noisy on a single field. The backend therefore couples two independent mechanisms: a binary-tier $P0 \cup P1$ *fusion* that always runs, and a five-rung *relaxation ladder* that fires only when a pool is empty.

P0 $\cup$ P1 fusion (default `p0_strategy = p0_union_p1`) runs unconditionally on every P0 plan. The backend issues the P0 topology query and a parallel `storey+type` query, then returns $P0 \cup (P1 \setminus P0)$ with topology-matched candidates ranked first and storey-mates appended as a safety net. The fusion is a deliberate binary-tier ranking, not a set intersection and not an empty-pool fallback, so GT is still recovered when one fingerprint field is mis-predicted.

The relaxation ladder fires inside a single P0 plan only when the current rung returns zero candidates, progressively dropping the most brittle filters: `full_fingerprint` $\rightarrow$ `no_position` $\rightarrow$ `topology_only` $\rightarrow$ `no_storey` $\rightarrow$ `relaxed` (drops the lowest-confidence triplet) $\rightarrow$ attribute fallback. Each demotion is recorded in `strategy_actually_used` and flags `fallback_triggered`, so the evaluation can separate clean P0 hits from relaxed ones. Fingerprint fields (`direction`, `object_subtype`, parsed `position_index`) are gated into the Cypher via `include_relation_fingerprint`; the oracle in Chapter 7 exercises the same levers to isolate which fields drive recall [24, 38].

For the running case AP_SK_294, executing the plan from Listing [Listing 6.3](#) produces the waterfall in Listing [Listing 6.4](#). The P0a rule fires with 9 raw-topology candidates; P0 ∪ P1 fusion then appends storey-mates to yield a 76-candidate shortlist. Lower-priority branches (P2, P4–P6, P8) are logged for diagnostic purposes only and do not alter the output.

Listing 6.4: Retrieval cascade trace for case AP_SK_294. P0 fires; remaining strategies are logged for diagnostic comparison.

```
[OK] P0 spatial_triplet : 9 raw -> 76 final (p0 U p1)
[--] P2 name_keyword   : empty pool
[--] P4 storey+type     : 76 candidates
[--] P5 storey_only     : 911 candidates
[--] P6 type_only       : 263 candidates
[--] P8 fallback        : 100 candidates (capped)
```

The 76-candidate shortlist is subsequently passed to the validation and handoff stage (Section [6.3.4](#)), which binds the selected GUID into a BCF 2.1 topic for downstream CDE review. Figure [Figure 6.6](#) illustrates the final disambiguation step, in which the domain user confirms the target element against the 3D viewer overlay.

3. Ranked candidates + confirmation

Showing 37 of 76 candidates in final backend order.

Rank	GT	Name	Type	Storey	GUID	Confirm
1		BALANS Fixed Single Window: BALANS 25M FLOOR (SH = 0): 780926	IfcWindow	Level 1	8xyFL4ySP5Tex0Ri78qDB1	☒ Approve
2		BALANS Fixed Single Window: BALANS 30M FLOOR (SH = 0): 778282	IfcWindow	Level 1	8xyFL4ySP5Tex0Ri78qDYL	☒ Approve
3		BALANS Fixed Single Window: BALANS 20M FLOOR (SH = 0): 916364	IfcWindow	Level 1	8HFJ4FEcn5UheCI1iy8Bbs	☒ Approve
4		BALANS Fixed Single Window: BALANS 15M FLOOR (SH = 0): 887538	IfcWindow	Level 1	3pFbansyD8K0HFtJHrTt\$5	☒ Approve

Figure 6.6. Human-in-the-loop confirmation for case AP_SK_294. The 76-candidate Priority 0 shortlist is highlighted in the 3D viewer; the domain user confirms the selected IfcWindow GUID before the BCF 2.1 handoff is finalized.

6.3.3 Evidence-Aware Graph-RAG Reranker

A Graph-RAG module serves as a post-retrieval layer, re-scoring candidates without altering pool composition. It executes lightweight Cypher queries to retrieve each candidate’s 1-hop graph neighborhood, then computes a set-intersection similarity against the extracted spatial constraints to promote structural matches.

6.3.4 Validation, Traceability, and Workflow Integration

The final stage of the pipeline converts retrieval outputs into traceable, schema-compliant workflow artifacts. Beyond merely identifying a candidate GUID, the system must validate the result, preserve the reasoning path, and package the output for downstream Common Data Environment (CDE) coordination.

6.3.4.1 Schema and Domain Validation

Implemented via the `src/rq2_schema/` module, the validation layer ensures the structural integrity of the final output. It performs standard JSON Schema validation, GUID existence checks, and basic attribute consistency checks (e.g., storey and class) against the loaded IFC model. While it does not execute full CORENET X regulatory validation [4, 14], this step ensures that the retrieved element is geometrically grounded and prevents hallucinated references from corrupting downstream workflows. Additionally, this module computes the completeness metrics utilized in the experimental evaluation (Chapter 7) to distinguish structurally valid extractions from semantically grounded ones.

6.3.4.2 Traceability, Explainability, and Handoff

To prevent the neuro-symbolic pipeline from functioning as an opaque black box, the system records a comprehensive execution trace for every query, detailing the extracted constraints, the selected query cascade strategy, and any fallback behavior. Because every retrieval is strictly tied to a concrete IFC GUID, frontline evidence remains explicitly linked to the authoritative BIM.

To facilitate seamless industry integration, the system utilizes a BIM Collaboration Format (BCF 2.1) handoff module. The `write_bcfzip()` function compiles the topic metadata, evidence snapshots, and selected GUIDs into a standards-compliant `.bcfzip` archive [9]:

```
<case_id>.bcfzip
  bcf.version
  <topic_guid>/
```

```
markup.bcf
viewpoints/
  viewpoint.bcfv
  snapshot.png
```

Crucially, the `viewpoint.bcfv` file embeds the retrieved `IfcGuid` within the `Components/Selection` block. Through validation, GUID grounding, and BCF packaging, the neuro-symbolic output ceases to be an isolated terminal result and becomes a traceable, interoperable artifact directly ingestible by conventional AEC coordination tools [9].

6.3.5 Demo and Observability

The interactive implementation of this stage is designed as an observability interface rather than only a presentation layer. It exposes the main stages of inference, including input assembly, constraint extraction, query planning, symbolic retrieval, graph inspection, and final visualization.

In the current interface, the user can upload site photos and floorplans, enter chat text, add 4D metadata, select the IFC model, and optionally attach a ground-truth case for visual comparison. After execution, the interface renders the pipeline in five visible stages: neuro-layer inference, constraint parsing with confidence gating, query-plan generation, Cypher execution, and candidate-pool inspection. A separate 3D viewer then highlights the returned pool and the selected or ground-truth GUID where available.

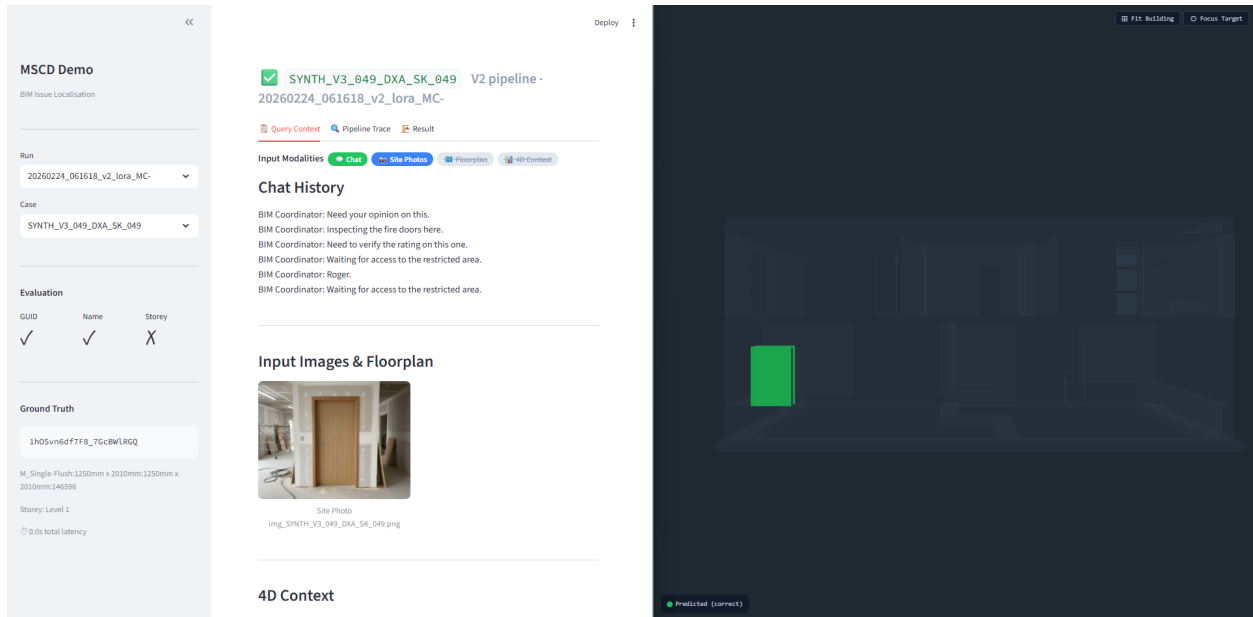


Figure 6.7. Demo and observability interface. The system exposes intermediate constraints, query plans, graph context, and grounded BIM candidates for inspection.

6.3.6 Summary

The implemented system operationalizes the interpreter middleware as a neuro-symbolic pipeline with explicit intermediate representations. The implementation can be summarized in three contributions. First, the neural front end converts mixed evidence into structured constraints rather than direct element predictions. Second, the symbolic layer grounds these constraints against a normalized and enriched IFC representation through deterministic planning and controlled fallback. Third, the validation and handoff layer preserves traceability from physical evidence to GUID-linked BIM outputs. Together, these components provide a practical bridge between ambiguous site observations and schema-ground digital records.

Chapter 7

Results and Discussion

This chapter presents the evaluation results of the neuro-symbolic IFC element retrieval system and discusses their implications for the thesis research questions. Section 7.1 reports the empirical results across four experiment groups. Section 7.2 synthesizes these findings through the lens of the two research questions.

7.1 Results

This section reports results across four experiment groups (Table Table 5.1). Group 1 compares the V1 agent-style prototype with the V2 structured pipeline. Group 2 isolates the symbolic retrieval ceiling through oracle and planner analyses. Group 3 evaluates the LoRA6 G-series models and Gemini across extraction quality and downstream GUID retrieval. Group 4 reports diagnostic studies on modality dependence, deterministic visual heuristics, and Graph-RAG reranking, which are interpreted further in Section 7.2.

7.1.1 Group 1: System Baselines

The first experiment group (design: Section 5.3.1) compares the earlier V1 agent-style system with the later V2 structured pipeline (system architecture: Section 6.1).

The main limitation of V1 was repeatability. Because tool choice, reasoning format, and intermediate search order remained open-ended, the pipeline was harder to audit, less stable under high spatial ambiguity, and more costly to run (Figure Figure 7.1). V1 was therefore useful as an exploratory prototype, but less suitable as the final retrieval architecture. The final system retains the agent mainly as an orchestration wrapper rather than as the primary grounding mechanism.

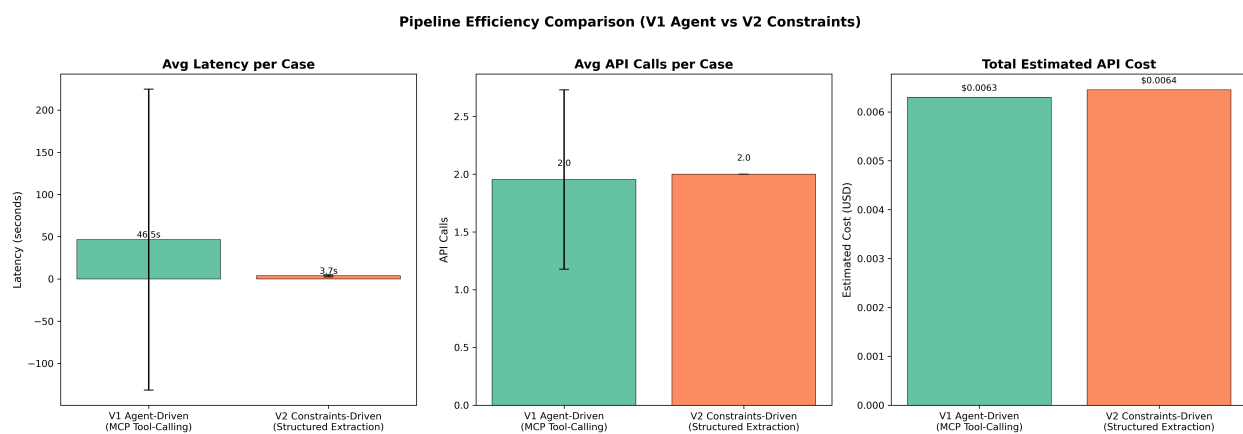


Figure 7.1. Operational efficiency comparison between V1 and V2. The structured pipeline reduces tool-call count and latency relative to the agent-driven baseline.

7.1.2 Group 2: Oracle Ceiling and Symbolic Limits

The second experiment group (design: Section 5.3.2) separates extraction quality from symbolic retrieval quality through three analyses: oracle ceiling performance, query planner strategy ablation, and fingerprint ladder analysis.

7.1.2.1 Ceiling Performance With Perfect Extraction

The oracle experiment evaluates the symbolic backend under perfect structured constraints, with the neural extractor bypassed. Figure 7.2 decomposes this ceiling into topology gain over the P1-only baseline, planner-redesign gain, and the remaining gap between oracle and learned extraction. The result shows that the symbolic layer can preserve the target element when supplied with correct constraints, while the remaining loss is primarily tied to extraction quality.

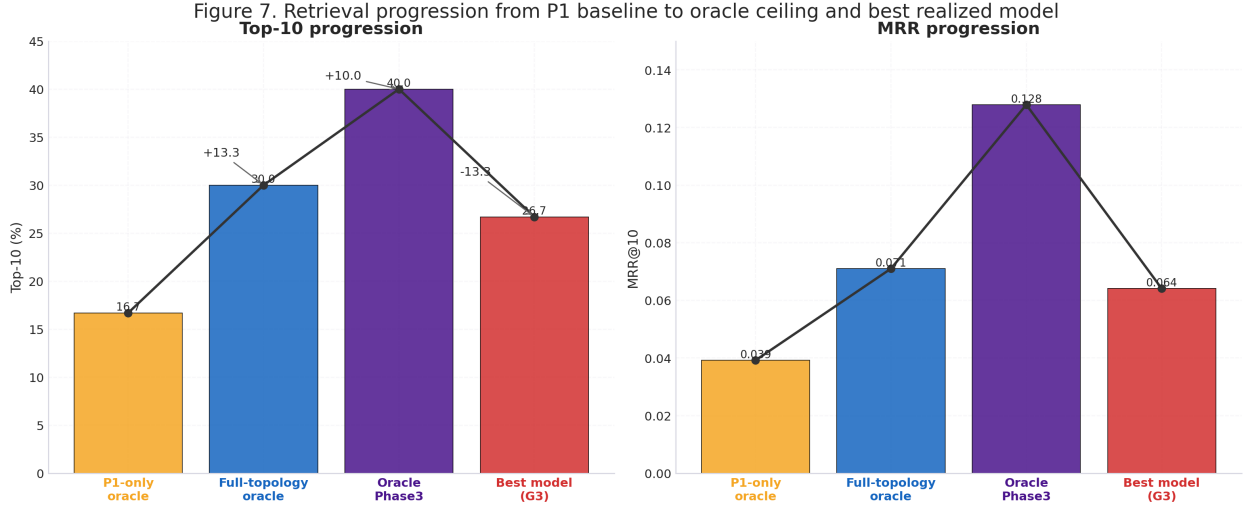


Figure 7.2. Retrieval progression from the P1 baseline to the oracle ceiling and the best realised model (G3). The four stages, P1-only oracle, full-topology oracle, the redesigned-planner oracle, and the best realised LoRA finetuned-VLM, are compared on Top-10 and MRR, isolating three distinct improvement sources: topology gain, planner redesign gain, and the remaining extraction gap.

7.1.2.2 Query Planner Strategy Ablation

The strategy ablation compares alternative ways of combining spatial P0 constraints with the P1 attribute baseline (priority cascade in Table 6.7; implementation in Section 6.3.1). As shown in Figure 7.3, p0Up1 is the safest executable strategy: it preserves recall while still allowing spatial constraints to improve ranking where topology is discriminative. Hard intersection over-prunes under noisy extraction, while P0-only retrieval is too brittle. The union strategy is therefore adopted as the canonical planner in the AP held-out downstream evaluation (Figure 7.5).

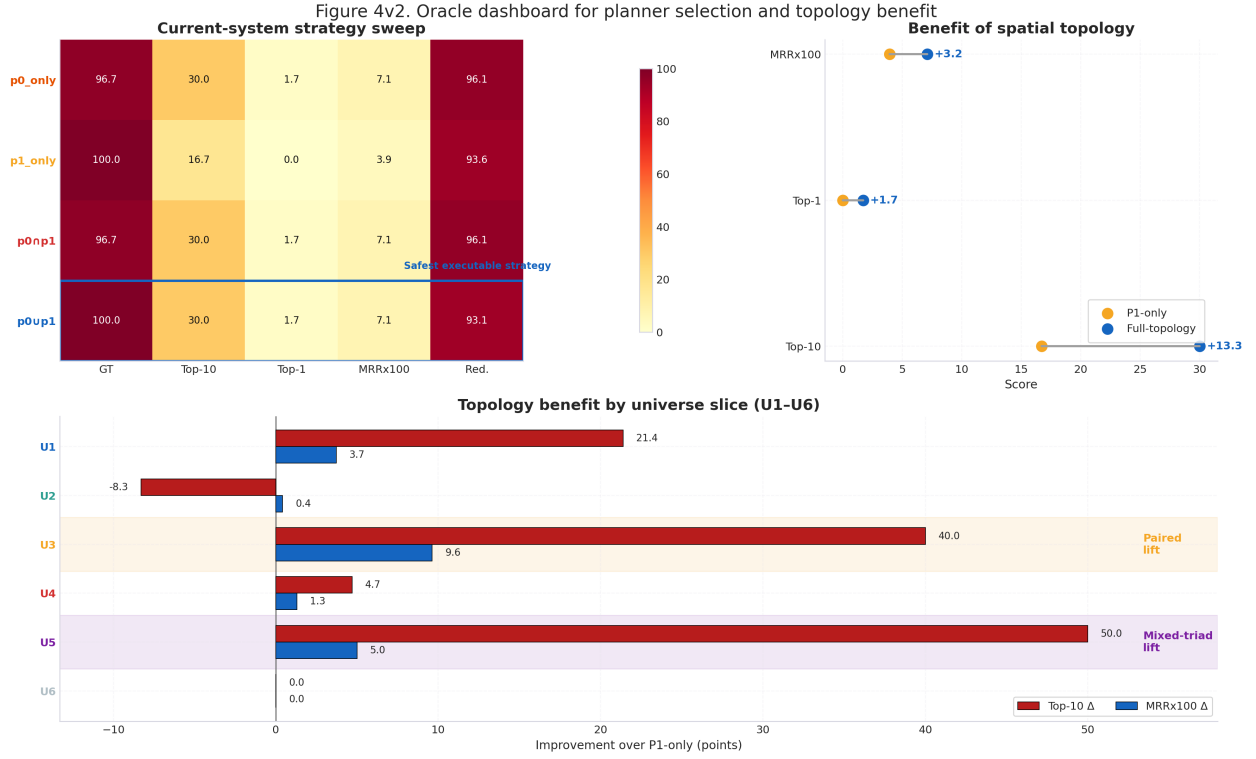


Figure 7.3. Oracle dashboard for planner selection and topology benefit. The upper-left matrix summarises the strategy search across the p0/p1 family; the upper-right panel compares P1-only against full-topology oracle on Top-10, Top-1, and MRR; the lower panel reports topology-sensitive gains by universe slice U1–U6. Together, these views confirm p0Up1 as the safest executable strategy and identify paired-opening and mixed-triad families as the clearest planner-opportunity cases.

7.1.2.3 Fingerprint Ladder Analysis

The **oracle ladder analysis** decomposes the symbolic ceiling into an eight-level progression (L0–L7), following the plot-script label convention in Table 7.1. L0–L4 estimate the theoretical pool and ranking behavior under progressively richer fingerprint matching, while L5–L7 report the corresponding live Cypher execution stages (Section 6.3.2). Coverage denotes the fraction of cases for which a fingerprint level is extractable from the graph.

The ladder shows that enriched graph value is concentrated in discriminative fingerprints rather than coarse topology alone. L2 provides limited compression because many elements share the same predicate-object pattern, such as `FILLS → IfcWall`. L3 is the dominant reduction step because it adds direction, subtype, material, distance, and connection degree. L4 shows the theoretical value of exact position-slot information, but only for the subset of cases where that signal is extractable. The current live planner therefore operates closer to

L2 than to the richer L3/L4 ceiling, making direction-and-subtype extraction and planner consumption the highest-value next implementation step.

Level	What it adds	Cov	Avg Pool (ideal)	Med Pool (live)	Ideal Top-10
L0	No Filter	100%	1257	1257	—
L1	Storey + IFC Class (P1 baseline)	100%	107.2	46	3.3%
L2	+ Predicate + Object (topology type)	100%	73.5	45	8.3%
L3	+ Direction + Subtype (fingerprint)	100%	24.4	9	58.3%
L4	+ Exact Position Slot (FILLS / NEXT_TO)	58.3%	0.74	1	100.0%
L5	+ Dimensions ± 50 mm	—	—	12	—
L6	Multi-anchor AND (2+ SRs, star pattern)	—	—	45	—
L7	p0Up1 (live default)	—	—	46	—

Table 7.1. Unified oracle ladder on the AP held-out benchmark ($n = 60$). L0–L4 follow the plot-script fingerprint convention; L5–L7 extend into the live Cypher execution layers. *Cov* is the fraction of cases where the level’s fingerprint is extractable from the graph, and *Avg Pool (ideal)* / *Ideal Top-10* are conditioned on that slice. *Med Pool (live)* is the median pool produced by the live `retrieval_backend.py` Cypher at each stage; L4–L6 operate on case subsets ($n \in \{35, 38, 33\}$), L0–L3 and L7 on all 60 cases. Em-dashes indicate columns not applicable at that level.

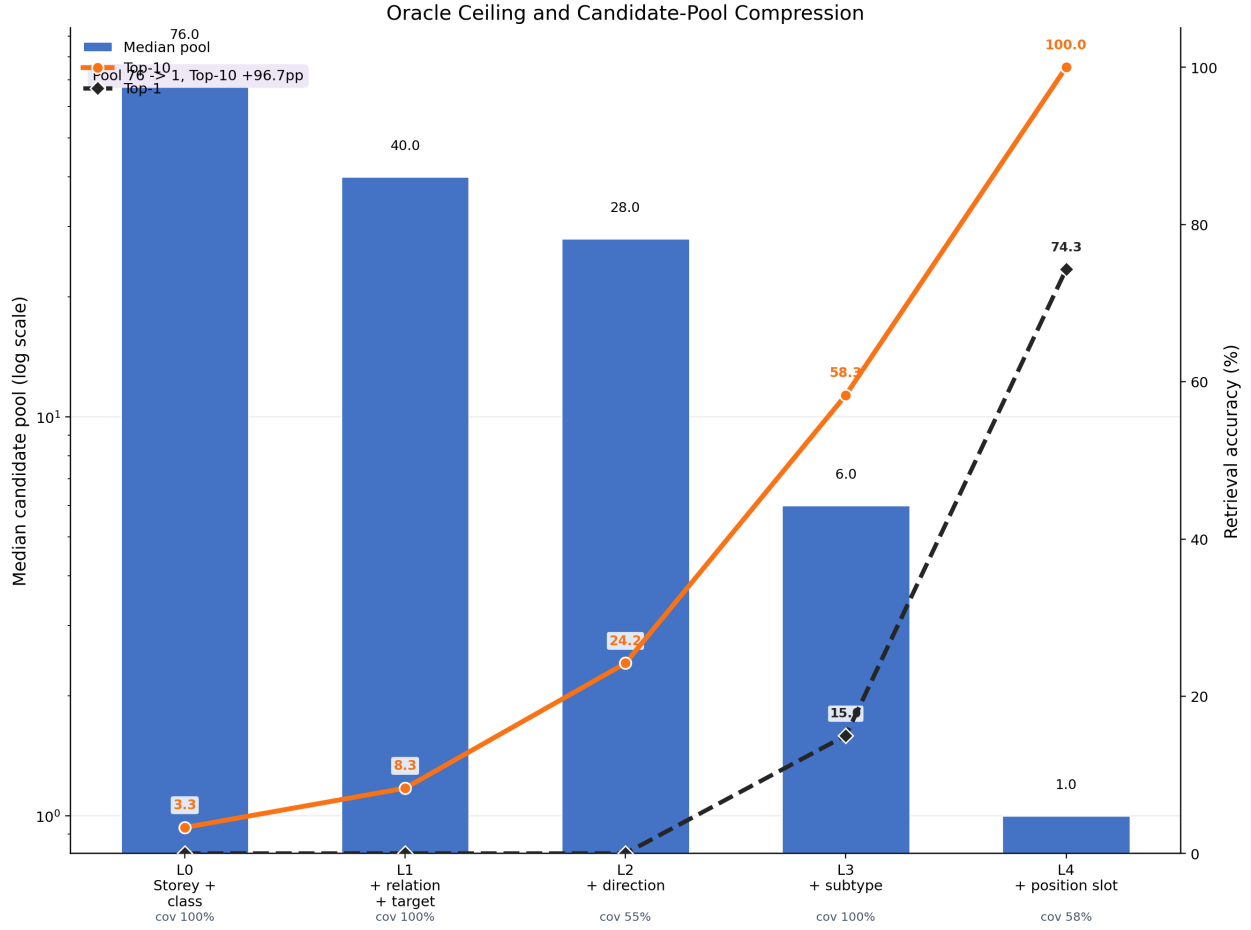


Figure 7.4. Oracle planner progression and fingerprint information loss side by side, decomposes the same ceiling into fingerprint levels, showing how direction- and subtype-aware consumption sharply shrink the pool and raise Top-10 and Top-1. This indicates highest theoretical ceiling available under richer fingerprint-aware planning.

7.1.3 Group 3: Dual-Track Evaluation of Vision-Language Models

7.1.3.1 End-to-End AP Benchmark Results

The third experiment group (Section 5.3.3) evaluates learned extractor variants on the AP held-out benchmark (Section 4.2.2.1) under the canonical p0Up1 planner. Figure 7.5 reports the downstream retrieval metrics, reports GT-in-Pool, Top-10, Top-1, and MRR@10 (metric definitions in Table 5.2) for the evaluated G-series LoRA6 models and Gemini (details of each model view at: Table 6.2). Table 7.2 gives the model-level comparison. Across configurations, the symbolic layer consistently reduces the search space, but downstream ranking remains sensitive to the quality and type of extracted spatial con-

straints.

A key finding is a three-way dissociation between extraction quality and downstream retrieval. G4 performs strongest on intermediate Hop-1 extraction but is not the strongest downstream retriever. G3 gives the most stable strict Top-10 retrieval, while the enriched position-context variants improve early-rank quality. This shows that model selection cannot be based on extraction accuracy alone; it must consider how extracted constraints interact with the symbolic planner.

The oracle-versus-model gap analysis (Figure [Figure 7.2](#)) confirms that, after planner redesign, the dominant remaining loss is attributable to extraction error rather than retrieval logic. The symbolic backend ceiling remains above all realised models, especially in cases where predicate, direction, or IFC class are incorrectly extracted.

7.1.3.2 Track A: Extraction Quality.

G8 preserves the core parsing strengths of G7 while improving multi-relation slot accuracy (Track-A metrics: Table ??). The gain is mainly in anchor-element identification rather than predicate recall. By contrast, G6 collapses on spatial extraction, confirming that topology-specific supervision is necessary for this task.

7.1.3.3 Track B: Downstream Retrieval.

Table [Table 7.2](#) summarises the full G-series and Gemini comparison on the AP held-out benchmark under the enriched graph.

Model	GT-Pool%	Top-1%	Top-10%	MRR@10	Med Pool	Graph
LoRA2 (FP-only) [†]	98.3%	0.0%	16.7%	0.0392	—	baseline
LoRA5-r32 (FP-only) [†]	100%	0.0%	21.7%	0.0407	—	baseline
G1 FullAug	100%	3.3%	23.3%	0.0645	76	baseline
G2 FullAug LowLR	100%	1.7%	20.0%	0.0524	76	baseline
G3 FullAug r32	100%	1.7%	26.7%	0.0641	76	baseline
G4 Ultimate	100%	0.0%	23.3%	0.0324	76	baseline
G6 Baseline	81.7%	1.7%	23.3%	0.0515	76	baseline
G7 PosCtx	100%	6.7%	26.7%	0.1015	76	enriched
G7 PosCtx + Graph-RAG	100%	1.7%	26.7%	0.0725	—	enriched + rerank
G8 PosCtx+Dim	100%	6.7%	30.0%	0.1104	76	enriched
Gemini v2	95.0%	1.7%	18.3%	0.0557	76	enriched
G9 + OpenCV + ResNet	98.3%	6.7%	26.7%	0.1020	—	enriched + band
G9 + OpenCV + ResNet + Graph-RAG	98.3%	3.3%	26.7%	0.0920	—	enriched + band + rerank

Table 7.2. Track B downstream retrieval on the AP held-out benchmark ($n = 60$), extended with predecessor LoRA adapters (LoRA2, LoRA5-r32), Phase 6 reranking, and G9 variants. G1–G6 use baseline-graph traces; G7, G8, Gemini v2, and the appended G9 variants use enriched-graph traces. [†]LoRA2 and LoRA5-r32 were evaluated under a floorplan-only input setup (no site photos or 4D metadata) and pre-date the enriched graph; rows kept for lineage context, not as apples-to-apples baselines. Dashes in Med Pool indicate that the corresponding summary artifact does not publish a median final-pool statistic.

Fine-tuned models overall outperform the weaker baselines (G6underfit model and gemini) in downstream retrieval. G8 shows the strongest combined Top-10 and MRR@10 performance, while the unchanged median pool indicates that the current p0Up1 planner often falls back to the P1 attribute pool. Improvements therefore appear more in ranking quality than in pool-size compression. G6 further shows that the enriched graph alone cannot compensate when topology extraction is weak.

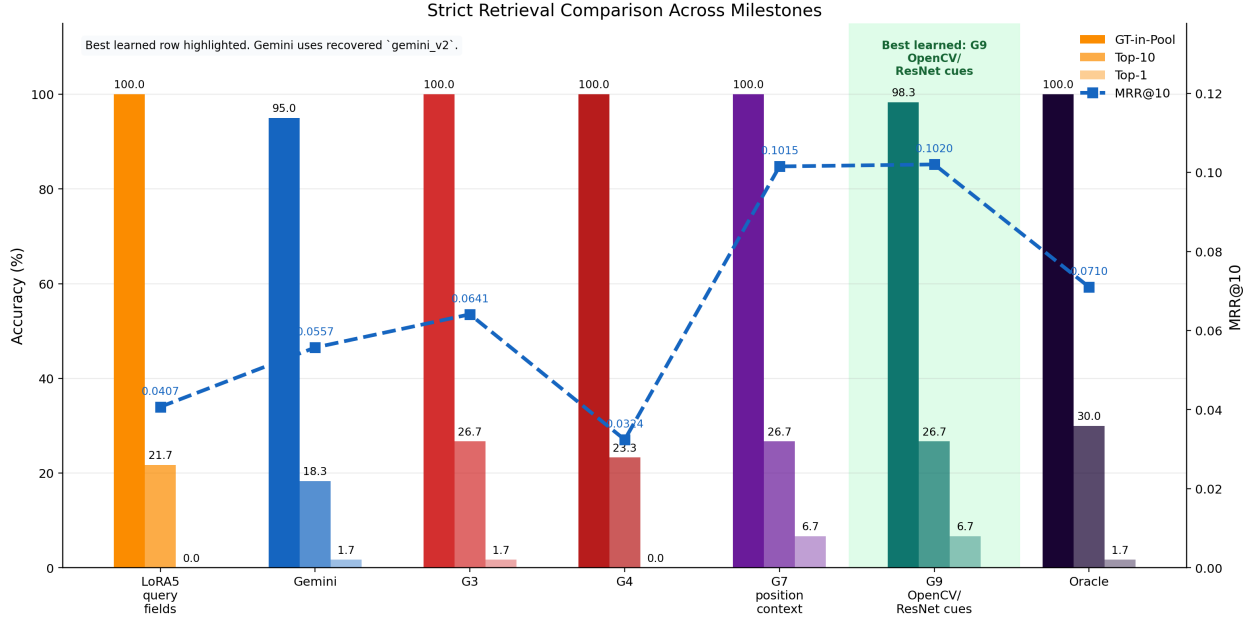


Figure 7.5. Track B strict downstream evaluation on the AP held-out benchmark ($n = 60$) with p0Up1. Grouped bars report GT-in-Pool, Top-10, and Top-1; overlaid lines show average final pool size and MRR@10. G3 achieves the strongest end-to-end ranking performance among the evaluated LoRA6 variants.

7.2 Discussion

The reported results are examined directly through the two thesis research questions. **RQ1** (Section 7.2.1) addresses how well the fine-tuned VLM grounds visual and textual evidence to produce spatial constraints under ambiguity, including the role of each modality and the limits of visual spatial reasoning. **RQ2** (Section 7.2.2) addresses how the symbolic Neo4j layer converts those constraints into ranked candidate pools, examining planner strategy, the oracle ceiling, and Graph-RAG reranking behaviour.

7.2.1 RQ1: Multimodal Grounding under Ambiguity

7.2.1.1 Fine-Tuning is Mandatory for Structured Spatial Extraction.

On the AP held-out benchmark ($n = 60$), fine-tuned G-series models outperform zero-shot Gemini v2 on every core extraction metric [Figure 7.5](#). The conclusion is unambiguous: domain fine-tuning is necessary for reliable structured extraction in AEC settings. Gemini v2’s near-zero direction accuracy (despite adequate type recognition at 63.3%) confirms that spatial direction grounding is not acquired from general pretraining and requires task-specific

supervision.

The performance advantage of fine-tuning is most pronounced at the extraction layer (Track A). After pool formation, the ranking advantage partially narrows—but G8 still leads Gemini on both GT retention and early-rank quality. The graph reranking results are evaluated separately and are not attributable to Gemini v2’s zero-shot capability.

7.2.1.2 Multimodal Fusion: Text as Anchor, Vision as Enhancer.

The evidence from the modality ablation indicates that spatial reasoning in the current system is not purely visual, but rather a successful multimodal fusion: the chat text acts as the necessary semantic anchor, while the visual evidence serves as a critical topological enhancer to reach peak accuracy.

Text is the load-bearing modality, and vision enhances spatial precision. Under text-only conditions, G8’s direction accuracy is only 28.6%; once the floorplan and site photos are introduced, it jumps to 82.1% — a >50-point gain (Figure [Figure 7.9](#)). Conversely, removing text entirely (visual-only FPSITE) collapses FILLS and ADJACENT_TO accuracy by tens of points, indicating that pure visual spatial reasoning fails rapidly without textual grounding. The system’s current spatial capability is therefore best characterised as *text-grounded topology extraction*.

Spatial topology improves retrieval over attribute-only baselines. The p0Up1 planner improves ranking over the text-only P1 baseline in 41 of 60 cases and places four ground-truth elements at rank 1 prior to any reranking (Figure [Figure 7.6](#)). Model with retrieving topology-discriminating predicate families (LoRA2-attribute only v.s. G-series finetuned VLM) reach higher accuracy and ranking score under full multimodal conditions(

Visual channels saturate quickly and lack dimensional precision. The floorplan combined with chat text already provides all available spatial signal; 4D point clouds add no measurable gain. Physical dimensions cannot be visually extracted: G8, the only model that attempts width and height predictions, achieves only single-digit accuracy ([Table 7.3](#)). Error analysis shows massive scale offsets, indicating that the rare successes reflect memorisation of standard architectural sizes (e.g., a 762mm door) rather than active visual measurement.

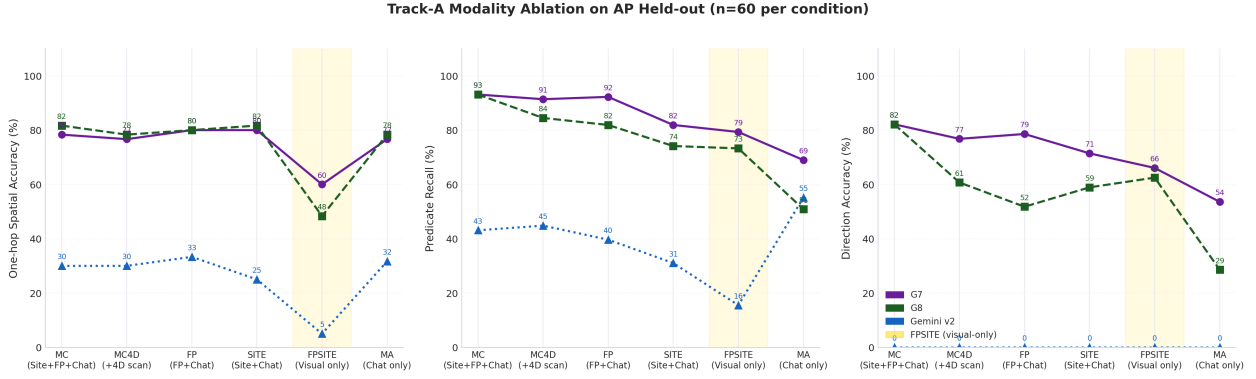


Figure 7.6. Track-A modality ablation on the AP held-out benchmark ($n = 60$ per condition) across six input conditions (MC through MA) for G7, G8, and Gemini. Left panel: one-hop spatial accuracy; centre panel: predicate recall; right panel: direction accuracy. The FPSITE column (visual-only, highlighted) shows the sharpest degradation across all models and all three metrics, confirming text as the load-bearing modality for spatial constraint extraction.

7.2.1.3 Decoupled Axes: Extraction vs. Retrieval.

Results demonstrate a clear dissociation between Track A (extraction) and Track B (retrieval) (5.3.3). While G4 leads in Hop-1 extraction accuracy, it performs poorest in strict Top-10 retrieval, suggesting that high-precision extraction prematurely prunes valid topological paths (Figure 7.7). In contrast, G3’s recall-heavy approach provides more exploitable candidates for the symbolic backend, securing the strongest Top-10 results. For early-rank quality, G8 excels ($\text{MRR@10} = 0.1104$), illustrating a trade-off between strict pool precision and ranking performance.

Field-level analysis (Table 7.3) reveals that while basic fields like *ifc_class* saturate at 100%, increased schema complexity induces a performance drop in peripheral attributes like *object_material*. Notably, G8’s superior retrieval despite lower material accuracy indicates that the VLM prioritizes higher-payoff spatial fingerprints over descriptive exactness, optimizing for discriminative power in the downstream task.

Field	G3	G4	G7	G8	Gemini	G9
<i>ifc_class</i> ($n = 60$)	100%	100%	100%	100%	63.3%	100%
<i>storey_name</i> ($n = 60$)	100%	100%	100%	100%	30.0%	100%
predicate slot-acc. ($n = 116$ SR)	86.2%	75.0%	80.2%	82.8%	43.1%	70.7%
<i>object_material</i> ($n = 116$ SR)	86.7%	80.0%	80.0%	78.3%	0.0%	70.0%
<i>position_context</i> exact ($n = 59$)	0.0%	1.7%	0.0%	8.5%	0.0%	15.3%
<i>position_context</i> type-match ($n = 59$)	35.6%	52.5%	35.6%	52.5%	20.3%	57.6%
<i>target_width_mm</i> correct ($n = 38$)	—	—	—	5.3%	—	—
<i>target_height_mm</i> correct ($n = 38$)	—	—	—	7.9%	—	—
<i>size_cluster</i> correct ($n = 38$)	—	—	—	—	—	31.6%

Table 7.3. Per-field extraction accuracy across the G-series, Gemini, and G9 on the AP held-out benchmark. Predicate and material are evaluated over all 116 GT SR slots using the same predicate-aligned field-level scorer; storey, class, and position_context are case-level. G9 deprecates direct mm regression and instead emits *size_cluster*, so width/height are absent from its output schema by design.

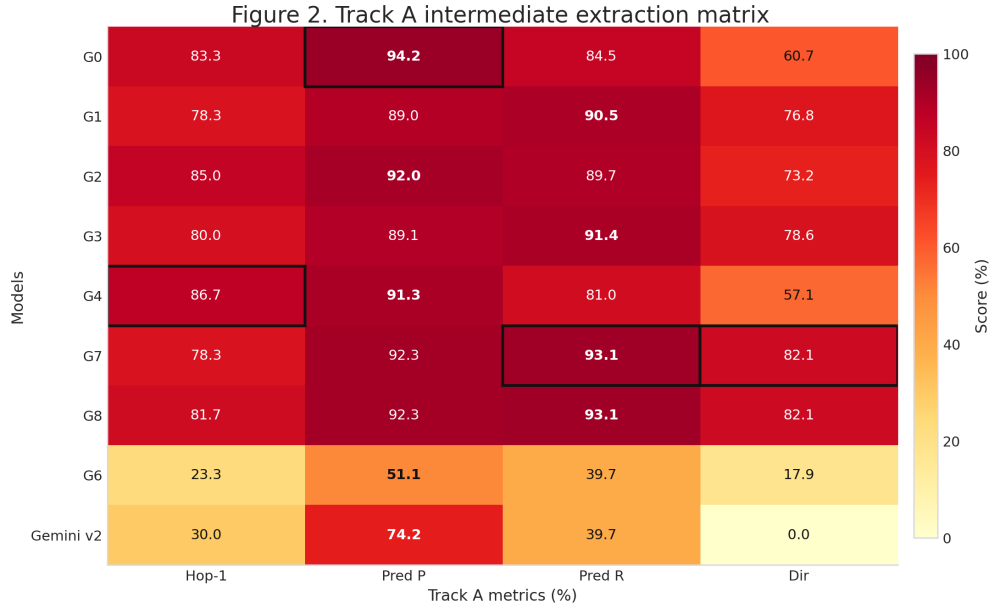


Figure 7.7. Track A intermediate extraction matrix on the AP held-out benchmark ($n = 60$). Rows correspond to evaluated models; columns report Hop-1 accuracy, predicate precision (Pred P), predicate recall (Pred R), and direction accuracy (Dir). G4 leads on Hop-1 only (86.7%); G7 and G8 lead on Pred P (92%), Pred R (93%), and Dir—the same models that lead in downstream Track B early-rank quality.

7.2.1.4 Deterministic Visual Heuristics Show Value

As mentioned in 6.2.3, deterministic visual heuristics and specialist classifiers are introduced to improve visual sub-tasks that require counting, ordinal position, or metric discrimination that weak on VLM, results from previous VLM model Table 7.3. Count and size predictions should not automatically become hard Cypher filters. Rather, they are most useful as supplementary evidence for candidate ordering and Graph-RAG reranking. This motivates the final reranking configuration, where OpenCV-derived ordinal cues are injected into the reranking prompt and improve early-rank retrieval compared with reranking without additional visual evidence.

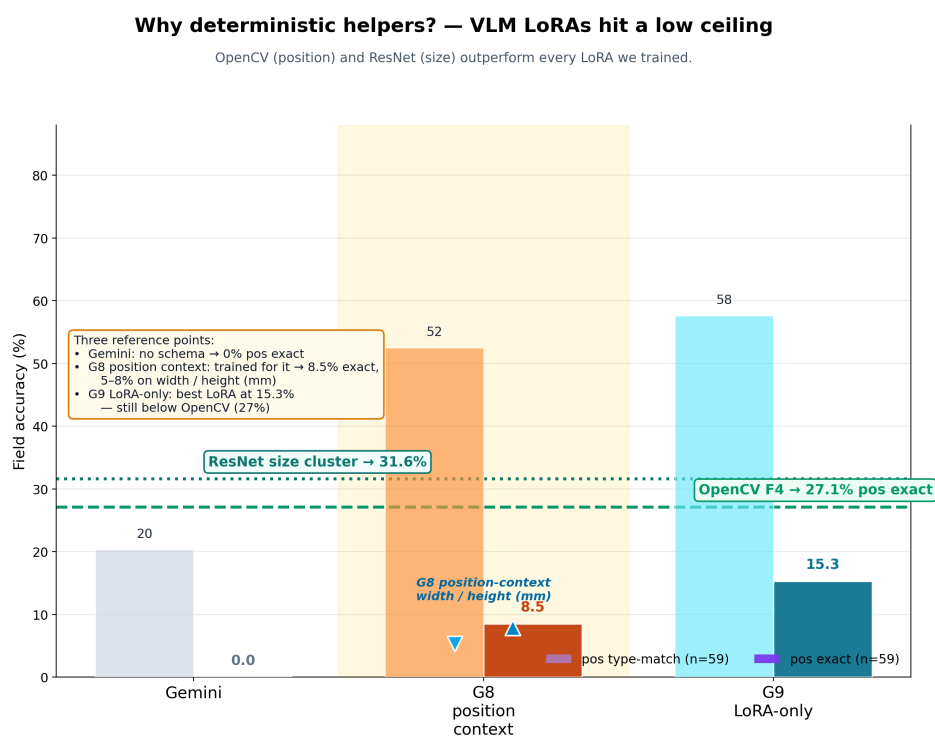


Figure 7.8. OpenCV for counting and ordinal position tasks, ResNet-based classification for element size groups outperform trained VLMs.

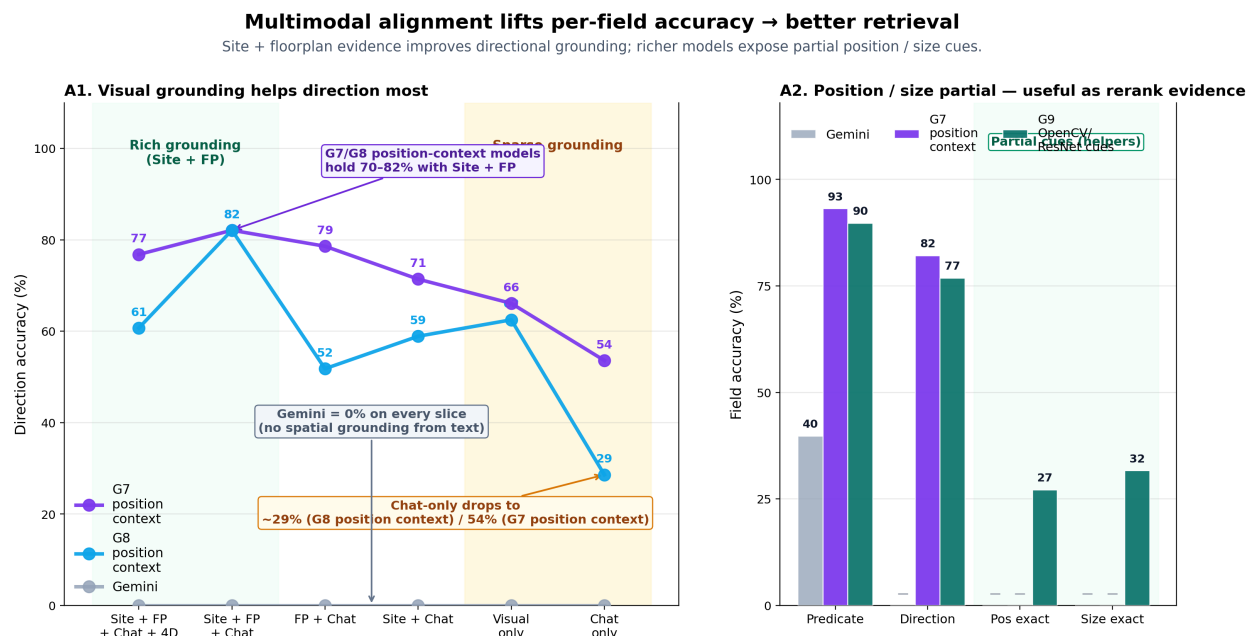


Figure 7.9. Multimodal visual evidence preserves directional grounding and exposes partial position/size cues, motivating heuristic visual reranking.

7.2.2 RQ2: Ontology-Driven Retrieval

7.2.2.1 The Symbolic Backend is Viable; Extraction is the True Bottleneck.

The oracle experiment confirms that the symbolic backend is structurally sound [Figure 7.2 Table 7.1](#). Under correct constraints, the graph reliably retains the ground-truth element (100% GT-in-Pool) and significantly compresses the search space. Specifically, advancing to an L3 fingerprint (Direction + Subtype added to L2’s Predicate + Object) increases the oracle Top-10 accuracy to 58.3%. Furthermore, L4 achieves a perfect Top-10 in 58% of cases where an exact Position Slot (FILLS / NEXT TO) is extractable.

Consequently, the residual loss in end-to-end retrieval stems from VLM extraction errors—primarily incorrect `ifc_class` gating or missing spatial predicates—rather than suboptimal retrieval logic. Crucially, the gap between current realized performance and the L3 oracle ceiling represents an extraction deficit, not a lack of planner features. Because the neuro-symbolic pipeline localizes errors to specific fields rather than treating them as opaque failures, the development roadmap is clear: the highest-value priority is improving VLM reliability in low-accuracy fields, such as subtype and position-context extraction, rather than redesigning the underlying graph planner [Table 7.3](#).

7.2.2.2 Query Strategy: The Union Planner and Efficient Spatial Relationship for Query

The strategy ablation shows that planner design has a direct effect on retrieval quality. Only the union strategy (p0Up1) preserves full GT-in-Pool while still benefiting from P0’s pool compression, and it accordingly serves as the canonical planner in the AP held-out downstream evaluation (Figure [Figure 7.3](#)).

This makes the planner a research contribution in its own right: the current evidence supports treating spatial relations as soft evidence for compression and reranking rather than as a strict filter.

Besides, from oracle ladder analysis result [Table 7.1](#). Author-added edges such as NEXT_TO and ADJACENT_TO become discriminative only when direction, subtype, and distance tolerance are consumed together. In contrast, IFC-native FILLS alone provides little reduction because many windows on the same floor share the same wall-filling relation.

The critical implication is that the enriched graph already contains useful discriminative structure, but the current learned extractor does not yet emit all of these fingerprints as stable structured JSON.

7.2.2.3 Ontology constrains the search; visual heuristics refine the final ordering through Graph-RAG reranking.

Figure [Figure 7.10](#) reports the Graph-RAG reranking comparison under two paired conditions on the G9 fine-tuned model: *full-topology* (with extracted spatial relations) and *P1-only* (storey + IFC class, no spatial relations). Reranking yields a significant improvement in both settings and rescues a number of edge cases that the symbolic stage alone cannot resolve.

The two stages play complementary roles. The symbolic P1 filter (IFC class + storey) establishes the deterministic baseline pool. The fine-tuned VLM then augments this pool with extracted spatial topology, while OpenCV and ResNet inject visual evidence, the ordinal slot position and size-band classification into the reranking prompt as discriminative cues. An earlier iteration surfaced this evidence as raw text without confidence weighting, which introduced noise; the final version fuses each evidence field with a per-candidate confidence score, producing measurable gains for both *P1 + rerank* and *full-topology + rerank*.

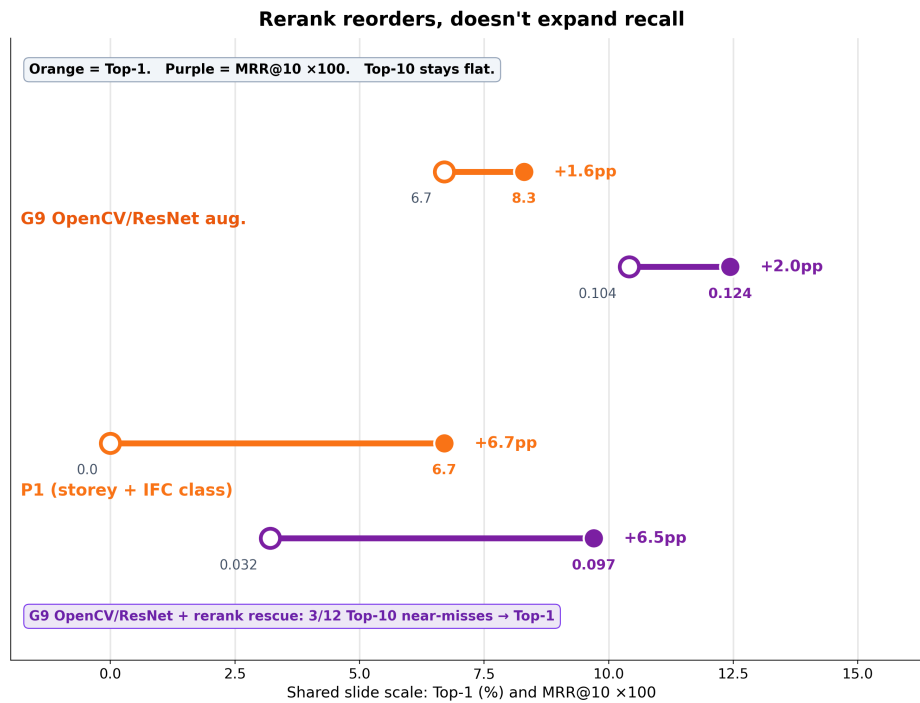


Figure 7.10. Final Graph-RAG prompt-mode comparison on the AP held-out benchmark. With OpenCV- and ResNet-derived ordinal evidence injected into the reranking prompt, single-shot Graph-RAG improves G9 on both Top-1 retrieval accuracy and MRR@10.

This pattern supports the broader design principle of *filter-rerank decoupling*: stable, high-precision fields should drive hard symbolic filtering, while uncertain or supplementary fields are best routed as soft reranking evidence, where a probabilistic ranker can absorb their noise without erasing recall.

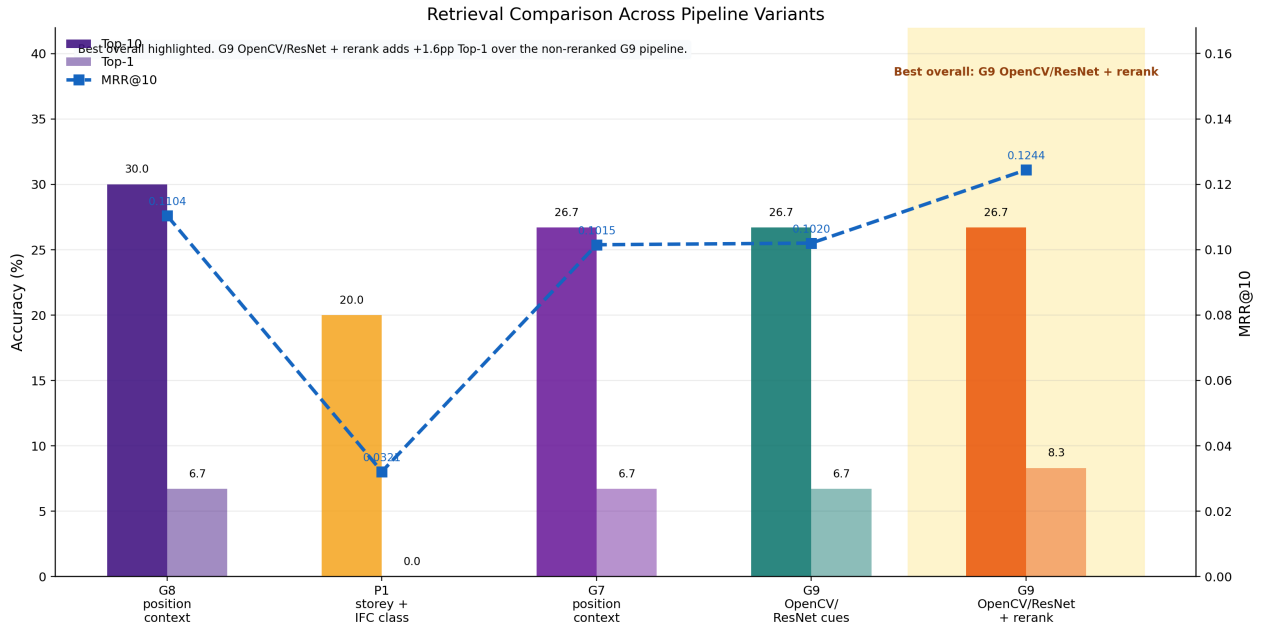


Figure 7.11. Effect of visual-evidence augmentation on downstream retrieval. Injecting OpenCV-derived ordinal evidence into the Graph-RAG prompt improves early-rank retrieval, supporting the interpretation that reranking improves when it receives new discriminative evidence not already consumed by the symbolic graph planner.

7.2.3 Summary

Taken together, the Chapter 7 results provide qualified support for both research questions. For **RQ1**, the experiments show that multimodal grounding is possible but not as pure visual recognition. Text provides the strongest semantic anchor, floorplan and site-image evidence improve topology extraction, and deterministic tools such as OpenCV are needed for ordinal cues that VLMs cannot reliably count. For **RQ2**, the oracle and ladder experiments show that the enriched IFC graph can preserve the correct element and compress the candidate pool when supplied with reliable constraints. However, the later diagnostic experiments show that richer fields must be routed carefully: stable fields can support hard filtering, while noisy fields such as position context, size cluster, and natural-language descriptors should be used as soft evidence or reranking signals. The main design lesson is therefore not simply to add more fields or a stronger reranker, but to assign each field to the part of the pipeline where its precision is sufficient.

Table 7.4. Summary of Chapter 7 evaluation findings and their relationship to the thesis research questions.

Chapter 7 section	Evaluation finding	Thesis interpretation	RQ link
Group 1: System baselines	V2 structured retrieval is more repeatable and efficient than the V1 agent-style prototype.	GUID-level grounding requires typed constraints and deterministic retrieval rather than open-ended agent reasoning alone.	RQ1, RQ2
Group 2: Oracle ceiling	The oracle planner reaches full GT-in-Pool under correct constraints and substantially reduces the candidate space.	The symbolic graph layer can preserve the correct IFC element when extracted constraints are reliable.	RQ2
Group 2: Fingerprint ladder	L3 fingerprints, especially direction, subtype, material, distance, and connection degree, provide the dominant compression gain.	The enriched IFC graph functions as an active retrieval structure, not merely as IFC storage.	RQ2
Group 2: Planner strategy	p0Up1 preserves recall better than hard intersection while retaining topology-based ranking value.	Spatial relations should guide retrieval as recall-preserving evidence rather than brittle exclusion filters.	RQ2
Group 3: LoRA vs Gemini	Fine-tuned G-series models outperform zero-shot Gemini on spatial fields required for graph retrieval.	AEC-specific spatial language benefits from domain supervision beyond general multimodal pre-training.	RQ1
Group 3: Track A / Track B	Higher extraction accuracy does not always produce better downstream retrieval.	The neuro-symbolic pipeline must be evaluated as coupled stages rather than as a single model score.	RQ1, RQ2
Group 4: Modality ablation	Text is the load-bearing modality; floorplan and site-image evidence improve topology; visual-only evidence collapses.	The system performs text-grounded topology extraction rather than pure visual spatial reasoning.	RQ1
Group 4: Deterministic visual heuristics	OpenCV-derived ordinal cues and preliminary size-band classification supplement VLM outputs, but remain auxiliary signals.	Some visual sub-tasks, especially counting and size estimation, are better handled as helper signals than as free-form VLM predictions.	RQ1

Table 7.4. Continued.

Chapter 7 section	Evaluation finding	Thesis interpretation	RQ link
Group 4: Filter-rerank decoupling	Noisy fields can harm retrieval when used as hard filters but remain useful as soft ranking evidence.	Stable fields should filter; uncertain fields should rank; low-confidence fields should trigger verification.	RQ2
Group 4: Graph-RAG reranking	Reranking hurts topology-filtered pools without new evidence, but improves early-rank retrieval when supplied with additional ordinal cues.	Graph-RAG is evidence-dependent: it helps only when it receives discriminative evidence not already consumed by the planner.	RQ2
Overall Chapter 7	The strongest behavior emerges when perception, symbolic retrieval, and reranking are assigned field-specific responsibilities.	The thesis contribution is a field-routed neuro-symbolic interpreter layer that makes ambiguity measurable, diagnosable, and reducible.	RQ1, RQ2

Chapter 8

Conclusion and Future Work

This thesis aims to bridge the gap between the structured reality of digital project data and the unstructured physical on-site data. By framing the traceability gap in Architecture, Engineering, and Construction (AEC) workflows, specifically the visual-geometric semantics gap exacerbated by high-entropy environments. This research contributes at conceptual, technical, and operational levels. Ultimately, this work establishes the foundation for a highly interoperable, streamlined data pipeline on the construction site, directly mitigating the vulnerabilities of semantic drift and context decay identified at the outset of this research.

8.1 Conceptual Contributions

This research demonstrates the computational possibility of swiftly mapping unstructured on-site evidence to structured architectural models, with answer the core research question: *How can AI act as an interpreter middleware to reliably align unstructured site evidence with project data in AEC workflows?* The evaluation of our neuro-symbolic pipeline provides answers through two sub-questions [1.4](#):

Research Question 1: Multimodal Grounding under Ambiguity The empirical results answer this partially in the affirmative, specifically under conditions that fuse text-rich context with spatial floorplans. Spatial grounding is highly learnable when textual, visual, and spatial cues are combined with deterministic heuristics like OpenCV for explicit counting and pre-clustering.

- **Enabling Zero-Shot Deficits:** Fine-tuning enables spatial grounding that foundational models cannot achieve; our G8 model reached 82% direction accuracy compared to a 0% baseline from zero-shot Gemini.
- **Multimodal Synergy:** Text and floorplans are highly complementary. Full modality (MC) achieved 82% accuracy versus 29% for text-only conditions (+53 percentage points).

- **Retrieval Impact:** Spatial topology materially improves candidate retrieval, successfully promoting the target to rank-1 in 41 out of 60 cases using the P0P1 strategy prior to any reranking, compared to 0 out of 60 for text-only baselines.
- **Scope and Limits:** The current paradigm remains text-dependent; relying exclusively on visual evidence (floorplan and on-site image, FPSITE) collapses the grounding accuracy to near-random performance.

Research Question 2: Ontology-Driven Retrieval The results conclusively validate the symbolic layer. By leveraging an enriched graph, the oracle bounds achieved 100% Ground-Truth-in-Pool retention. Furthermore, integrating topological fingerprints (direction, subtype, and distance constraints) drastically compresses candidate pools, collapsing the median pool size from 24 to 1 at the L4 fingerprint level. Spatial queries vastly outperform attribute-only filters. Crucially, this confirms that the remaining operational ceiling is bound by neural extraction accuracy, not by missing graph logic.

8.2 Technical Contributions

This thesis contributes a functional prototype architecture that successfully acts as the proposed interpreter middleware, proving three core technical capabilities:

- **Multimodal Visual Grounding:** The deployment of a fine-tuned multimodal extractor capable of transforming ambiguous site images, floorplan patches, and chat logs into strictly typed spatial constraints. These constraints natively translate into executable IFC graph queries, a task where zero-shot generative models fail entirely.
- **Ontology-Driven Symbolic Retrieval:** The development of an IFC Parse and Enrich Engine that structurally supports deterministic retrieval. This engine not only guarantees 100% ground-truth retention under oracle conditions but also drives the deterministic generation of synthetic training data directly from raw IFC files.
- **Decomposable Neuro-Symbolic Architecture:** By explicitly decoupling probabilistic extraction from deterministic graph retrieval, the architecture eliminates opaque black-box failures, enabling highly targeted failure diagnosis and preserving a traceable chain of reasoning from site evidence to BIM GUID.

8.3 Workflow Impact: The Interpreter Middleware in Action

By combining multimodal visual grounding with ontology-driven symbolic retrieval, the proposed middleware reduces hallucination risk and changes how unstructured site evidence can be managed in AEC workflows. Rather than allowing a model to directly invent or select building elements, the system constrains every candidate to an existing IFC entity and exposes the retrieval result for human verification.

To clarify the operational logic of the proposed system, a storyboard is provided to illustrate the end-to-end workflow from multimodal input to structured BIM output. It visualizes how user intent is progressively interpreted, grounded, and translated into executable modeling actions through the multi-layer architecture.



Figure 8.1. Storyboard illustrating the end-to-end workflow of the proposed interpreter layer, showing the transformation from multimodal input to structured BIM actions through sequential perception, reasoning, and execution stages.

At the micro-level, the system accelerates site workflows by making ad-hoc evidence traceable. Ambiguous photos, informal annotations, and site messages can be anchored to the authoritative Building Information Model (BIM) or cloud-based Industry Foundation Classes (IFC) repository, reducing the manual interpretation bottleneck described in Section 1.1.2.

This also helps mitigate context decay during personnel transitions, because the record is no longer only a loose image or chat message, but a model-linked reference that can support schema-compliant reporting.

At the macro-level, mapping isolated site evidence to spatial BIM context enables project teams to aggregate and visualize site dynamics, including defect density, spatial trade conflicts, repeated inspection issues, and regulatory blind spots. In this sense, the interpreter middleware does not replace professional judgment. Instead, it changes the role of the human reviewer from manually searching and reconstructing context to verifying, correcting, and escalating model-linked evidence.

Operationally, this middleware can affect site data management through the following workflows.

1. Human-in-the-Loop Triage and Verification In environments with high attribute entropy and repeated building elements, the system acts as a search-space compressor. By reducing the candidate pool from thousands of elements to a concentrated recommendation list, Construction Managers (CMs) can rapidly verify the context and select the correct GUID. This shifts the CM’s role from conducting tedious manual database searches to performing targeted verification.

2. Continuous Supervised Fine-Tuning The results also suggest that fine-tuning should be viewed as part of the deployment workflow, not only as a one-time modeling choice. At the beginning of a project, a lightweight fine-tuned model may already be useful for shortlist generation. As more confirmations and corrections are collected during daily work, the same model can be updated to better fit the project’s vocabulary, drawing conventions, and recurring spatial patterns.

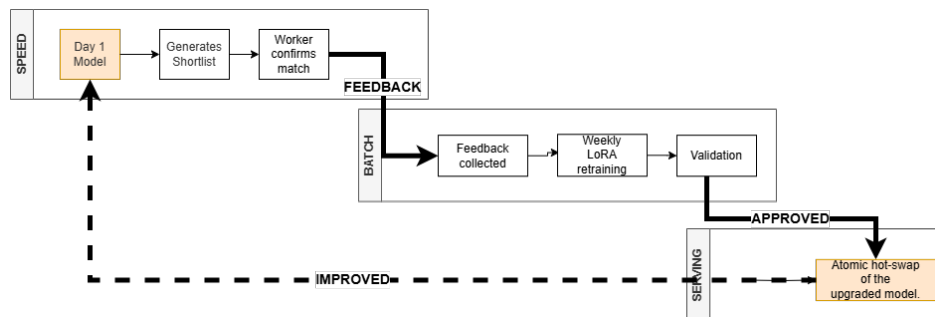


Figure 8.2. Lambda-style deployment view focused on the feedback loop.

Worker feedback becomes a direct training signal (Figure [Figure 6.6](#)). Accepting a retrieved candidate provides a positive example, while rejecting it and selecting another element provides corrective supervision. Over time, this creates a feedback loop in which the system improves through use while keeping human verification in control of model-linked decisions.

3. Automated Cross-Stakeholder Communication As system reliability improves, the middleware can be integrated into stakeholder communication channels. For example, a frontline worker might submit an informal note: *"On Level 1, the window beside the door at this location is damaged and needs extra manpower."* The interpreter can process this unstructured evidence, apply topological reasoning, and append a deterministic identifier, such as `target_guid: "1ebX2H7F12su$wtPbeYATe"`. The back-office coordinator then receives a more actionable record for scheduling, inspection, or issue tracking, without manually reconstructing the link between the message and the BIM element.

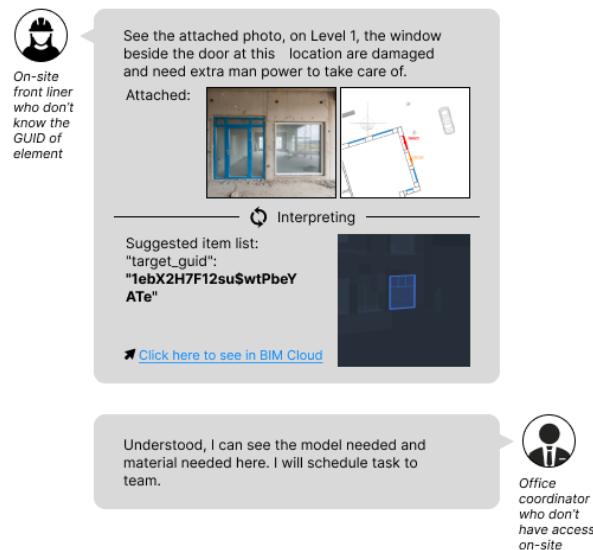


Figure 8.3. Sample of automated cross-stakeholder communication.

4. Natural Language Search and Retrieval and Highlevel Analytics Once this semantic link is established, stakeholders across the Common Data Environment (CDE) can use natural language to search BIM elements or retrieve historically grounded site evidence. A query can refer to a human-readable description, while the backend retrieves model-linked records through their associated GUIDs, spatial relations, and issue history.

At the project level, continuous ingestion of grounded site data enables analytics and visualization over accumulated site evidence. The following figures illustrate how the interpreter layer can support operational memory, spatial traceability, and model-linked dashboards.

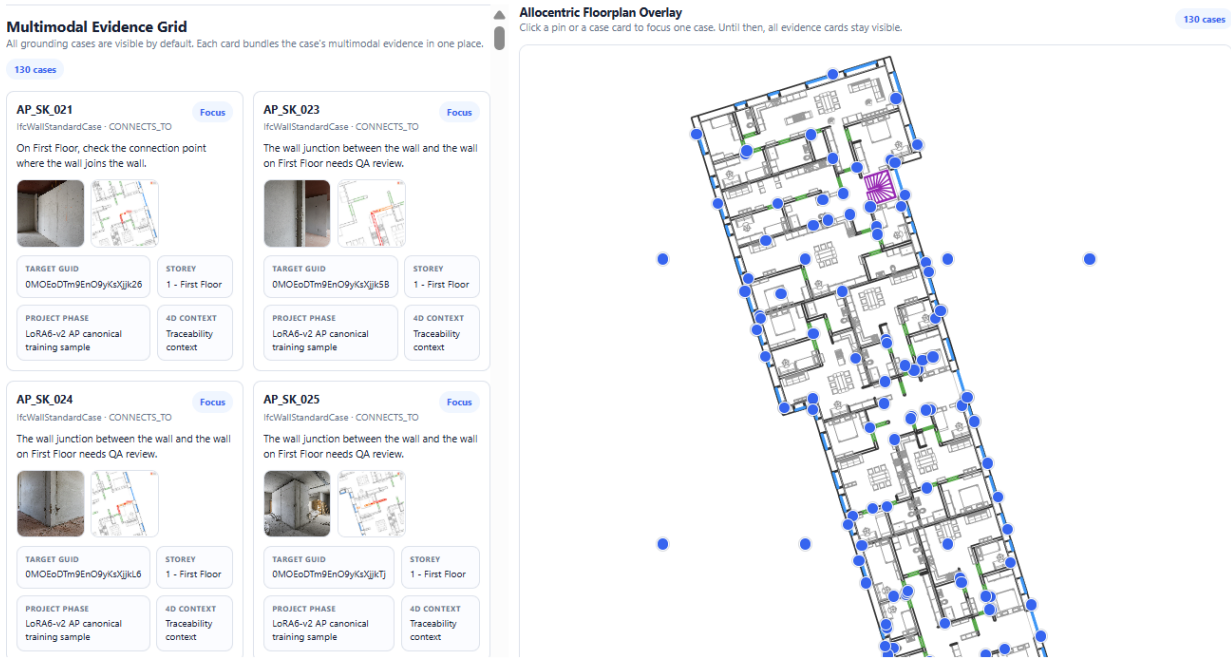


Figure 8.4. Visualizing the neuro-symbolic bridge: grounding egocentric site evidence within allocentric BIM topology.

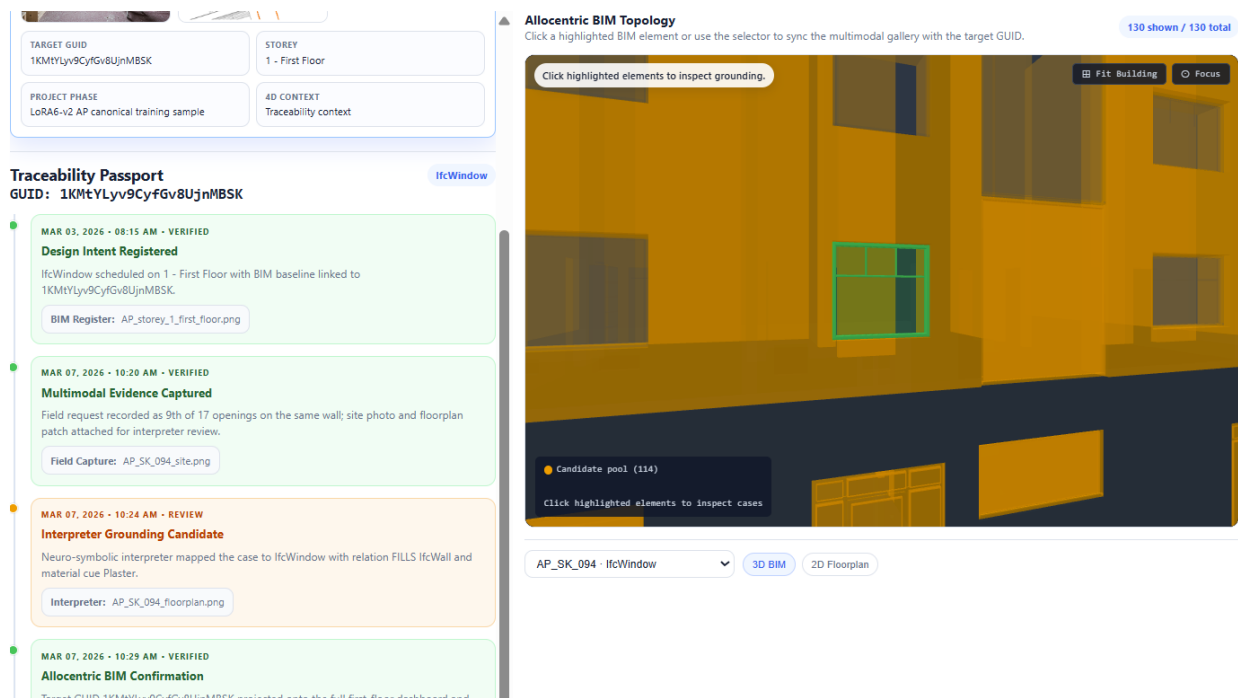


Figure 8.5. Maintaining institutional memory and mitigating context decay through element-level traceability.



Figure 8.6. High-resolution traceability heatmaps aggregating element-level data to reveal macro-trends, such as defect density, trade conflicts, and regulatory blind spots, that remain difficult to detect in traditional siloed reporting.

8.4 Future Work and Vision

8.4.1 Limitations

This thesis has several limitations that define the scope of the current prototype and the direction of future work. First, the evaluation is still primarily based on **synthetic and semi-synthetic scenarios**. Although these scenarios are derived from IFC geometry and controlled graph structures, they do not fully capture the visual noise, occlusion, lighting variation, informal annotation style, and distribution shift present in live construction-site images. As a result, **cross-modal generalization** from generated evidence to real on-site evidence remains an open challenge.

To mitigate this limitation, the prototype introduces floorplan patches as an additional input modality. These patches function as a **semi-structured bridge between visual evidence and symbolic retrieval**. They are not part of the IFC schema and are not directly traceable to a specific BIM element ID, but they are generated deterministically from the model geometry and therefore provide a reliable spatial reference. This makes them useful as a stabilizing modality, even though they do not yet solve the broader problem of real-world visual generalization.

Second, the neuro-symbolic layers are currently **connected through orchestration rather than fully fused at the model layer**. The neural extractor produces structured constraints, and the symbolic graph backend executes retrieval using those constraints. However, the symbolic layer is not yet differentiable or directly learnable from retrieval outcomes. For example, the current system does not include a regression or calibration model that links per-field extraction confidence to downstream graph-retrieval behaviour. Future work should therefore explore more integrated learning mechanisms in which symbolic retrieval outcomes can inform field weighting, uncertainty estimation, and planner policy.

8.4.2 Future work

Near-term future work will focus on transitioning from synthetic sandboxes to live deployments. This includes extending the training datasets with real project traces, deepening the integration with live CDE APIs, refining uncertainty and escalation policies for the query planner, and exploring the system’s generalization capabilities beyond high-entropy PPVC environments to other construction archetypes.

Ultimately, this thesis serves as the foundational logic for a broader future vision: toward continuous tracking.

Deployment Architecture: Continuous Supervised Fine-Tuning. Although the continuous supervised fine-tuning loop has not yet been deployed as a production system, the current prototype with data mining pipeline 4 and learnable VLM Table 7.2 provides a strong foundation for it, mitigating day-one model cold start issues and leveraging worker feedback data. 8.3

Dynamic 4D Arrangement: The current standard of relying on weekly coordination meetings creates inherent schedule lag. By ingesting and grounding live, unstructured site data, Construction Managers can rearrange schedules and logistical sequencing dynamically, responding to the real-time physical status of the site.

Autonomous On-Site Tracking: The manual upload of chat logs and site photos is only a transitional phase. The ultimate objective is the deployment of an autonomous onsite camera network that utilizes this exact neuro-symbolic interpreter logic. Such a system will continuously track, mathematically ground, and topologically analyze site progress 24/7, bridging the physical-digital divide without requiring human intervention.

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Appendix A

Demo Code

The source code for this thesis, including the prototype implementation and experimental scripts, is available at https://github.com/hyChia88/mscd_demo.git.

Appendix B

Extractor Prompts and Message Templates

This appendix reproduces the prompts used by the neural extraction layer described in Section 6.2. All LoRA and prompt-only extractors share a single authoritative source (`mscd_demo/prompts/constraints_extraction.yaml`) so that training, inference, and baseline comparisons remain aligned. Three artifacts are listed below: the LoRA system prompt (C.1), the prompt-only system prompt used for Gemini/GPT baselines (C.2), and the generic user-message template with a concrete instantiation for case AP_SK_294 (C.3).

B.1 LoRA System Prompt (`lora_system_g9`)

The following single-paragraph system prompt is used both at LoRA training time and at inference. Keeping the inference prompt byte-identical to the training prompt avoids a distribution shift between the fine-tuned adapter and the runtime extractor. The prompt is short by design—rich schema guidance is instead taught through paired training examples—so that the fine-tuned model learns the output contract from data rather than from prompt wording.

Listing B.5: `lora_system_g9` system prompt used by the LoRA-based extractor.

```
You are a construction site assistant that extracts IFC search
constraints from multimodal evidence. Use the floorplan and site
photo to reason about storey, element type, position context,
size, and spatial relations. Output valid JSON only with fields
storey_name, ifc_class, space_name, target_name_keyword,
position_context, size_cluster, and spatial_relations.
```

```
If the user prompt includes an [OpenCV Counting] block, copy its
position/total numbers into position_context (format
"Nth of M openings on the same wall"); do not re-guess.
```

```
For IfcWindow or IfcDoor targets, set size_cluster to a label such
as window_M_1480x1380 or door_S_760x2030.
```

Each spatial relation must use predicate/object_type and may include object_subtype, direction, object_material, confidence. Only include direction, object_subtype, position_context, or size_cluster when supported by evidence. Do not guess. Return JSON only.

B.2 Prompt-Only System Prompt (Gemini Baselines)

Zero-shot baselines such as Gemini 2.5 cannot rely on fine-tuned weights to internalize the output contract, so they receive a longer schema-corrected prompt that explicitly enumerates fields, enums, and rules. This is the prompt behind the “Gemini AP” row in Table [Table 6.2](#); it is deliberately never used for LoRA inference to keep training/inference parity intact.

Listing B.6: prompt_only_system prompt used for Gemini/GPT zero-shot baselines.

You are a construction site assistant that extracts search constraints from multimodal inputs (photos, floorplans, chat messages, and metadata). Given the conversation and any attached images, extract structured JSON constraints to identify the BIM element being discussed.

Output ONLY valid JSON with these fields:

```
{
  "storey_name": "exact floor name or null",
  "ifc_class": "IfcWall|IfcWindow|IfcDoor|IfcSlab|IfcStair|... or
    null",
  "space_name": "room/space name or null",
  "target_name_keyword": "equipment brand/ID/unique name or null",
  "position_context": "'3rd of 17 openings on the same wall' or null
    ",
  "spatial_relations": [
    {"predicate": "FILLS|ADJACENT_TO|CONTINUOUS|NEXT_TO|CONNECTS_TO
      ",
      "object_type": "IfcClass",
      "object_subtype": "string or null",
      "direction": "left|right|null",
      "object_material": "material or null",
      "confidence": 0.0-1.0}
```

```
]
}
```

Rules:

- storey_name must match exact IFC storey names (e.g., '1 - First Floor', 'Level 1', '-1 - Garage').
- ifc_class must use the Ifc prefix (e.g., 'IfcWindow' not 'window').
- space_name: extract if user says 'in the kitchen', 'room 601', 'the bathroom window'; null otherwise. Do NOT copy storey_name into space_name space_name is a room, not a floor.
- target_name_keyword: extract equipment IDs like 'AHU-03', 'Daikin unit', 'Fire Pump FP-01'; null for generic types like 'window' or 'wall'.
- spatial_relations: when a floorplan is provided, extract structural relationships visible in the plan.

predicate	uppercase; one of FILLS, ADJACENT_TO, CONTINUOUS, NEXT_TO, CONNECTS_TO.
object_type	Ifc-prefixed class.
direction	only for NEXT_TO when left/right ordering is supported by the evidence.
confidence	0.0-1.0 reflecting certainty.

 If no floorplan is provided, spatial_relations = [].
- position_context: use slot language only when the floorplan / topology supports it; otherwise null.
- Be conservative: use null if uncertain.

B.3 User-Message Template

All extractors (LoRA and prompt-only) consume the same user-message envelope, which bundles the runtime inputs defined in Section 6.1: a site photo, a rendered floorplan patch, optional 4D task metadata, the chat transcript, and an optional [OpenCV Counting] block supplying the deterministic ordinal count from Section 6.2 (OpenCV augmentation). The generic schema is shown first, followed by the concrete instantiation for the running case AP_SK_294.

B.4 Generic Template

Listing B.7: Generic user-message template consumed by every extractor variant.

```
[
{"role": "system", "content": "<system prompt      C.1 or C.2>"},
{"role": "user",   "content": [
  {"type": "image", "image": "file:///<path>/site_photo.png"},
  {"type": "image", "image": "file:///<path>/floorplan_patch.png"},
  },
{"type": "text",   "text":
  "[4D Task Status] <task name / phase>\n"
  "[Project Phase] <project-wide milestone>\n"
  "[Chat Log]\n"
  "  <role>: <message>\n"
  "  ... \n"
  "[OpenCV Counting] <Nth of M openings on the same wall>  #
    optional\n"
  "[Query] identify the BIM element\n"
  "Extract the search constraints as JSON."
  }
  ]}
]
```

B.5 Instantiation for Case AP_SK_294

The following message is the exact envelope delivered to the extractor for the running case. The [OpenCV Counting] hint is supplied by the deterministic heuristic of Section 6.2; without that block the extractor would either omit `position_context` or re-guess, which would break the downstream `wall_position_index` filter used in Listing Listing 6.3.

Listing B.8: Concrete user message for case AP_SK_294. The corresponding extracted JSON appears in Listing Listing 6.2.

```
[
{"role": "system", "content": "<lora_system_g9      Listing C.1>"},
{"role": "user",   "content": [
  {"type": "image", "image": "file:///.../AP_SK_294_site.png"},
  {"type": "image", "image": "file:///.../AP_SK_294_scale_M.png"},
  {"type": "text",   "text":
    "[4D Task Status] Window Installation\n"
    "[Project Phase] Facade Works\n"
  }
  ]}
]
```

```
"[Chat Log]\n"
"  supervisor: On Level 1, check the window beside the door at
    this location.\n"
"[OpenCV Counting] 2nd of 3 openings on the same wall\n"
"[Query] identify the BIM element\n"
"Extract the search constraints as JSON."
}
]}
```